Mathematical and Physical Modeling of Entangled Photon Propagation in Optical Fibers

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Chapter 1

General Framework

1.1 Purpose and Scope

The purpose of this work is to establish a coherent framework for the mathematical modeling and numerical simulation of polarization-entangled photon pairs propagating through optical fibers.
The central objective is to organize the physical assumptions, mathematical formalism, and simulation strategy into a reproducible structure, so that each computational step can be traced back to a precise theoretical element. In this way, the simulator is not developed as an isolated numerical tool, but as a direct operational translation of the underlying quantum model.

The modeling approach adopted throughout this work is intentionally progressive. Simplified analytically tractable models are introduced first, and are then extended toward increasingly realistic descriptions incorporating statistical effects, decoherence, and effective noisy evolution.
This strategy enables a controlled connection between physical interpretation, analytical predictions, and numerical validation.

1.2 Theoretical Formalism and Notation

The system is described within the framework of quantum information theory. The relevant mathematical structures include finite-dimensional Hilbert spaces, quantum states, tensor products, unitary operators, density operators, and projective measurements.
Elements of classical optics are introduced only when required to justify the physical origin of the effective quantum transformations.

The notation and formal tools are aligned, as far as possible, with the framework developed in *Quantum Computation and Quantum Information* by Nielsen and Chuang [1].

Within this formalism:

- single subsystems are represented by vectors in $\mathbb{C}^2$,
- composite systems are described through tensor-product spaces,

- evolutions are represented by linear operators, unitary in the ideal case,
- measurements are described through Hermitian operators.

All states and operators are represented as vectors and matrices over $\mathbb{C}$, with dimensions determined by the tensor-product structure of the system.

1.3 Physical Scenario and Experimental Setting

The physical setting considered in this work is the following. s A spontaneous parametric down-conversion (SPDC) source generates pairs of entangled photons, ideally prepared in a Bell state.
The polarization degree of freedom is used to encode the qubit associated with each photon. The two photons are injected into distinct optical paths, denoted A and B, and propagate through independent optical fibers.

Each fiber acts on the polarization degree of freedom of the corresponding photon. In a general description, this action can be represented by a unitary operator acting on a single-qubit Hilbert space.
However, motivated by experimental constraints and by the limited degree of control available on the complete polarization transformation, the model is initially restricted to a simplified effective description.

After compensation of global polarization rotations, the dominant residual effect can be modeled as a relative phase shift between the horizontal and vertical polarization components.
In the qubit representation, this corresponds to a rotation around the computational Z-axis of the Bloch sphere.

The objective at this stage is therefore not to reproduce the full electromagnetic propagation inside optical fibers, but rather to construct an effective quantum model capable of capturing the relevant impact of fiber propagation on entanglement and measurable correlations.

1.4 Computational and Modeling Architecture

The computational framework developed in this work follows directly from the theoretical structure introduced in the previous sections. The simulator is designed as a modular numerical realization of the underlying quantum model, preserving a direct correspondence between physical concepts and computational components.

At the numerical level, pure states are represented as complex vectors, mixed states as density matrices, and physical evolutions and observables as matrix operators. This representation enables a unified treatment of unitary dynamics, statistical mixtures, expectation values, and effective noisy evolution within the same computational framework.

The simulator architecture is organized around distinct functional layers, including state preparation, operator construction, local evolution, observable evaluation, quantum channels, and experiment-specific routines. Each layer corresponds to a well-defined theoretical block and can therefore be validated independently before being integrated into more advanced models.

Particular emphasis is placed on progressive model refinement. Simplified analytically tractable configurations are used as validation baselines before introducing increasingly realistic descriptions involving stochastic phase evolution, ensemble averaging, and non-unitary effects.

At the first modeling stage, unnecessary optical detail is neglected and each fiber is described exclusively through its effective action on polarization. This leads to a local unitary representation of the form

$$U_{AB} = U_A \otimes U_B \tag{1.1}$$

where $U_A, U_B \in \mathbb{C}^{2\times 2}$ are unitary operators.

This abstraction defines the fundamental interface between the physical system and the simulator, and will later allow the introduction of more refined models, including stochastic phase evolution, effective decoherence, and non-unitary quantum channels.

The overall objective of the computational framework is therefore not only to reproduce numerical results, but also to maintain analytical transparency, reproducibility, and consistency between theoretical predictions and numerical implementation throughout the progressive development of the model.

Chapter 2

Modeling Strategy

2.1 Progressive Modeling Philosophy

2.1.1 Progressive Modeling Approach

The modeling approach follows a progressive refinement strategy, in which the system is described through successive levels of approximation of increasing physical complexity.
Each stage introduces additional physical effects while remaining consistent with the previous ones. This ensures analytical transparency and preserves continuity between simplified and refined models.

From a theoretical perspective, this approach isolates the contribution of individual mechanisms. From a computational perspective, it enables a layered simulator structure, where each refinement corresponds to a well-defined extension of the existing codebase.

2.1.2 Hierarchy of Modeling Levels

The modeling workflow is structured as a sequence of progressively refined descriptions:

Ideal Bell State
↓
Local Phase Model (Z-rotations)
↓
General Local Unitary Transformations
↓
Mixed-State Description (Density Operators)
↓
Noisy Quantum Channels
↓
Effective Fiber Models

More explicitly:

- **Level 1: Minimal phase model** Pure, maximally entangled state with fiber action reduced to a relative phase. This level captures the analytical dependence of correlation observables on a single parameter.

- **Level 2: General unitary dynamics** Arbitrary polarization rotations described by local operators $U_A \otimes U_B$.
- **Level 3: Mixed-state description** Density operators used to represent statistical mixtures and loss of coherence.
- **Level 4: Noisy quantum channels** Effective completely positive trace-preserving (CPTP) maps modeling depolarization and decoherence.
- **Level 5: Effective fiber models** Physically motivated descriptions, such as concatenations of random unitary segments or stochastic polarization evolution.

This hierarchy defines the conceptual roadmap of the present work.
The initial levels remain analytically tractable and allow direct validation, while the later levels progressively increase physical realism and connect to experimentally relevant propagation effects.

2.2 Baseline Quantum Model

2.2.1 Pure-State Description

At the first stage, the bipartite system is described as a pure state in $\mathbb{C}^2 \otimes \mathbb{C}^2$, and fiber propagation is modeled through local unitary transformations.
Particular emphasis is placed on relative phase shifts induced by the fibers.

The main objective is to analyze how these phase shifts affect measurable two-photon correlations.
The relevant observables are operators of the form

$$\sigma_i \otimes \sigma_j, \qquad i, j \in \{x, y, z\},$$

as well as operators corresponding to arbitrary measurement directions on the Bloch sphere.

These operators play a central role because they describe joint local measurements performed independently on the two subsystems.
More precisely, the operator σ_i acts on subsystem A, while σ_j acts on subsystem B. Their tensor product therefore represents a correlation measurement between the outcomes obtained along two specified measurement directions.

The corresponding expectation values
$\langle \sigma_i \otimes \sigma_j \rangle$
quantify the statistical correlations between local measurements performed on the two photons.
These quantities contain direct information about the nonlocal structure of the bipartite quantum state and therefore constitute one of the fundamental signatures of entanglement.

For polarization-entangled states, these observables are particularly important because they are directly connected to experimentally accessible coincidence measurements.
In the computational basis, the Pauli operators correspond to measurements along specific polarization bases:

- σ_z: horizontal/vertical basis,
- σ_x: diagonal/antidiagonal basis,
- σ_y: circular polarization basis.

Consequently, expectation values of the form

$$\langle \sigma_i \otimes \sigma_j \rangle$$

can be interpreted as experimentally measurable polarization correlations between the two photons.

Beyond their direct physical interpretation, these correlators also provide the building blocks for more advanced mathematical descriptions introduced later in this work.
In particular:

- the diagonal correlations $\langle \sigma_i \otimes \sigma_i \rangle$ characterize the persistence of polarization correlations under fiber propagation,
- the complete set of correlators defines the correlation tensor representation of the bipartite state,
- combinations of these expectation values enter directly into Bell inequalities and CHSH nonlocality tests.

This baseline model therefore constitutes the simplest regime in which entanglement is preserved and can be probed exclusively through correlation observables.
It provides a controlled setting for deriving analytical expressions and validating the numerical implementation.

This regime establishes the minimal framework required to connect fiber-induced phase evolution with measurable two-photon correlations, while simultaneously introducing the correlation structures that will later be generalized to mixed states, arbitrary measurement axes, and non-unitary evolution.

2.2.2 Extension to Mixed-State Formalism

At the next level, the model incorporates statistical effects and decoherence.
In this regime, the pure-state description is no longer sufficient and must be replaced by a density-operator formalism.

This extension enables the description of:

- reduced states of individual photons,
- loss of coherence due to unresolved degrees of freedom,

- effective non-unitary dynamics arising from environmental interactions or statistical averaging.

Within this framework, physically relevant quantities such as purity, fidelity, and concurrence can be defined and computed, enabling a quantitative characterization of mixedness, coherence, and entanglement (see Chapter 4).

A central mathematical operation in the density-operator formalism is the partial trace, which allows the description of individual subsystems starting from the global bipartite state.
For a composite system described by the density operator ρ_{AB}, the reduced state of subsystem A is obtained by tracing over subsystem B:

$$\rho_A = \mathrm{Tr}_B(\rho_{AB}) \tag{2.1}$$

Analogously, the reduced state of subsystem B is

$$\rho_B = \mathrm{Tr}_A(\rho_{AB}). \tag{2.2}$$

The reduced density operators contain all information accessible through local measurements performed on the corresponding subsystems.

For maximally entangled Bell states such as $|\Psi^+\rangle$, the reduced states are maximally mixed:

$$\rho_A = \rho_B = \frac{1}{2}I. \tag{2.3}$$

This result constitutes one of the fundamental signatures of bipartite entanglement. Although the global state is pure, each subsystem individually admits a mixed-state description, implying that no local measurement can fully reconstruct the global quantum state.
The information characterizing the entangled system is therefore encoded nonlocally in the correlations between the two subsystems.

This connection between entanglement, reduced-state mixedness, and nonlocal correlations plays a central role throughout the present work, both in the theoretical analysis and in the interpretation of the numerical simulations developed in later chapters.

Expectation values in the density-operator formalism are computed as

$$\langle O\rangle = \mathrm{Tr}(\rho O) \tag{2.4}$$

which generalizes the pure-state expression $\langle\psi|O|\psi\rangle$.

The density-operator formalism therefore provides the natural framework for describing ensemble averaging, decoherence, and effective non-unitary dynamics arising during fiber propagation.

2.3 Computational Architecture and Modeling Roadmap

2.3.1 Simulator Design and Architecture

The simulator is designed as a modular implementation of the theoretical model.

The initial modules include:

- Bell-state generation,
- local unitary propagation,
- correlation evaluation.

More advanced modules introduce:

- quantum channels,
- reduced density operators,
- entanglement measures,
- experimentally relevant observables such as Bell parameters and Stokes-like quantities.

The theoretical structure is directly reflected in the computational architecture, ensuring consistency, traceability, and extensibility.

The implementation distinguishes clearly between reusable computational kernels and higher-level experiment scripts, allowing the simulator to evolve into a structured and reusable research tool.

2.3.2 Implementation Strategy

The simulator is designed as a layered computational framework mirroring the progressive hierarchy of the theoretical model.

At the initial level, the implementation operates on pure two-qubit states represented as complex vectors in $\mathbb{C}^4$.
The corresponding modules include:

- Bell-state generation,
- local unitary evolution,
- expectation-value evaluation,
- correlation computation.

The extension to mixed-state dynamics introduces:

- density-matrix representations,
- partial-trace operations,

- observable evaluation in the density formalism,
- state metrics such as purity, fidelity, and concurrence.

More advanced layers incorporate:

- quantum channels,
- ensemble-based evolution,
- stochastic propagation models,
- experimentally motivated observables.

At the current stage, this structure is reflected in the repository organization:

- `Core`: state vectors, operators, expectation values, and correlation routines;
- `Models`: physical propagation models;
- `Utils`: validation routines and auxiliary tools;
- `Experiments`: scripts implementing complete simulation studies.

This modular architecture preserves analytical traceability while enabling incremental extensions of the physical model without structural modifications of previously validated components.

Chapter 3

Physical and Mathematical Representation

3.1 Quantum System and Hilbert Space Structure

The physical system considered consists of polarization-entangled photon pairs generated by spontaneous parametric down-conversion (SPDC). The two photons propagate along distinct optical paths, denoted A and B, forming a bipartite quantum system.

In this work, the qubit is encoded in the polarization degree of freedom:

$$|H\rangle \equiv |0\rangle, \qquad |V\rangle \equiv |1\rangle. \tag{3.1}$$

Accordingly, the computational basis of the bipartite system is

$$\mathcal{B}_{AB} = \{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}. \tag{3.2}$$

In polarization notation, this corresponds to

$$\{|H\rangle_A|H\rangle_B, |H\rangle_A|V\rangle_B, |V\rangle_A|H\rangle_B, |V\rangle_A|V\rangle_B\}.$$

The Hilbert space of the total system is therefore

$$\mathcal{H}_{AB} = \mathcal{H}_A \otimes \mathcal{H}_B \cong \mathbb{C}^2 \otimes \mathbb{C}^2. \tag{3.3}$$

Since each subsystem is two-dimensional, the composite Hilbert space has dimension 4.
As a consequence:

- pure states are represented by complex vectors of length 4,
- operators acting on the full bipartite system are represented by 4×4 matrices.

This construction specifies the concrete realization of the abstract quantum-information framework introduced previously. The fixed computational basis in Eq. (3.2) will be used consistently throughout the theoretical development, the numerical implementation, and the validation experiments.

3.2 Bell-State Initialization

As a reference initial condition, this work considers the Bell state

$$|\Psi^+\rangle = \frac{1}{\sqrt{2}} \left(|H\rangle_A|V\rangle_B + |V\rangle_A|H\rangle_B\right). \tag{3.4}$$

Using the computational-basis convention introduced in Eq. (3.1), the same state can be written as

$$|\Psi^+\rangle = \frac{1}{\sqrt{2}} \left(|01\rangle + |10\rangle\right). \tag{3.5}$$

In the ordered computational basis $\mathcal{B}_{AB}$, this state is represented by the column vector

$$|\Psi^+\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ 1 \\ 1 \\ 0 \end{pmatrix}. \tag{3.6}$$

The state $|\Psi^+\rangle$ is maximally entangled and provides the natural input for the baseline version of the simulator. It also serves as the reference state with respect to which later quantities such as fidelity and Bell-inequality violation may be evaluated.

The Bell state provides both a clean analytical benchmark and a physically relevant entangled resource, enabling direct interpretation of fiber-induced effects in terms of coherence, two-photon correlations, and nonlocality.

For completeness, the four Bell states are

$$|\Psi^\pm\rangle = \frac{1}{\sqrt{2}} \left(|01\rangle \pm |10\rangle\right), \qquad |\Phi^\pm\rangle = \frac{1}{\sqrt{2}} \left(|00\rangle \pm |11\rangle\right). \tag{3.7}$$

Among these, $|\Psi^+\rangle$ is used as the default reference input for the propagation models considered in this work.

3.3 Fiber Propagation as Local Unitary Evolution

Each photon propagates through its own optical fiber. At the level of the polarization qubit, the action of each fiber is described by a linear operator acting locally on the corresponding single-photon Hilbert space. In the ideal lossless description, this operator is unitary.

If U_A and U_B denote the single-qubit operators associated with the two fibers, the evolution of an initial two-photon state is

$$|\psi_{\text{out}}\rangle = (U_A \otimes U_B)\, |\psi_{\text{in}}\rangle. \tag{3.8}$$

This expression follows directly from the tensor-product structure of composite quantum systems and represents the natural bipartite extension of single-qubit evolution. It provides the fundamental starting point for subsequent refinements, including random unitary perturbations, concatenated fiber sections, and effective channel descriptions.

In this framework, the two fibers act independently on the two polarization subsystems, and the total evolution is obtained by tensor composition of the corresponding local transformations. This description remains valid as long as the fibers do not induce any direct interaction between the two photons.

The local-unitary model is therefore the first dynamical layer of the thesis: it captures deterministic polarization transformations while preserving the purity of the global two-photon state.

3.4 Reduced Phase Model

Although a general fiber transformation may involve arbitrary polarization rotations, birefringence, and mode-coupling effects, the first approximation adopted here is simpler and directly motivated by the experimental control scheme.
After partial compensation of polarization drifts, the dominant residual effect is modeled as a relative phase shift between the horizontal and vertical polarization components. In the qubit representation, this corresponds to an effective rotation around the computational Z-axis of the Bloch sphere.

At the single-qubit level, this is represented by the diagonal unitary

$$U(\theta) = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\theta} \end{pmatrix}, \tag{3.9}$$

defined up to an irrelevant global phase.

If the two fibers induce phases θ_A and θ_B, the total transformation is

$$U_A \otimes U_B = U(\theta_A) \otimes U(\theta_B). \tag{3.10}$$

Applying this operator to the reference Bell state $|\Psi^+\rangle$ yields

$$|\psi(\theta_A, \theta_B)\rangle = \frac{1}{\sqrt{2}} \left(e^{i\theta_B} |01\rangle + e^{i\theta_A} |10\rangle \right). \tag{3.11}$$

The phase accumulated by each computational component depends on which subsystem carries the logical state $|1\rangle$. In particular, the component $|01\rangle$ acquires the phase generated by subsystem B, while the component $|10\rangle$ acquires the phase generated by subsystem A.

Factoring out the global phase $e^{i\theta_B}$, which has no observable effect, gives the reduced one-parameter state

$$|\psi(\theta)\rangle = \frac{1}{\sqrt{2}} \left(|01\rangle + e^{i\theta} |10\rangle \right), \qquad \theta = \theta_A - \theta_B. \tag{3.12}$$

Consequently, only the relative phase difference between the two arms has physical significance. Any common phase contribution acts as a global phase factor and therefore does not modify observable quantities.

With the convention adopted throughout this work, the reduced state in Eq. (3.12) can be interpreted equivalently as arising from a local phase rotation acting only on subsystem A,

$$U_A(\theta) = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\theta} \end{pmatrix}, \qquad U_B = I, \tag{3.13}$$

so that

$$(U_A \otimes I)|\Psi^+\rangle = \frac{1}{\sqrt{2}} \left(|01\rangle + e^{i\theta}|10\rangle \right). \tag{3.14}$$

This convention is adopted for consistency across the analytical derivations and numerical implementation developed throughout the present work.

Interpretation. In this reduced description, the physically relevant fiber action is encoded in a single parameter: the relative phase θ, determined by propagation conditions such as birefringence, path length, and environmental fluctuations.
The model isolates the dominant contribution affecting two-photon correlations while preserving a pure-state description. As a consequence, entanglement remains maximal and all observable variations arise from phase-induced modifications of quantum coherence.

Role in the modeling hierarchy. The reduced phase model constitutes the first nontrivial propagation model in the hierarchy. It provides an analytically tractable reference case for validating the simulator and serves as the baseline for subsequent extensions, including general unitary dynamics, ensemble descriptions, and effective non-unitary evolutions.

3.5 Correlation Tensor Representation

Bipartite correlations are described through the Pauli-operator basis

$$\sigma_i \otimes \sigma_j, \qquad i, j \in \{x, y, z\},$$

constructed from the single-qubit Pauli matrices $\sigma_x, \sigma_y, \sigma_z$. It is also useful to introduce $\sigma_0 = I$, so that local terms and correlations can be treated within a single notation.

For a two-qubit state ρ_{AB}, the expectation values

$$T_{ij} = \mathrm{Tr}\left(\rho_{AB}\, \sigma_i \otimes \sigma_j\right), \qquad i, j \in \{x, y, z\}, \tag{3.15}$$

define the components of the correlation tensor $T \in \mathbb{R}^{3\times 3}$.

More generally, the state admits the Pauli decomposition

$$\rho_{AB} = \frac{1}{4} \sum_{i,j=0}^{3} S_{ij}\, \sigma_i \otimes \sigma_j, \qquad S_{ij} = \mathrm{Tr}\left(\rho_{AB}\, \sigma_i \otimes \sigma_j\right), \tag{3.16}$$

where S_{ij} are generalized Stokes parameters. This representation is commonly used in quantum-state tomography of two-qubit photonic systems, where the density operator is reconstructed from measured expectation values of tensor products of Pauli operators [2].
In this representation:

- S_{i0} and S_{0j} encode the local Bloch vectors of the two subsystems,
- the block S_{ij}, with $i, j \in \{x, y, z\}$, corresponds to the correlation tensor T_{ij}.

The orthogonality relation

$$\mathrm{Tr}\left[(\sigma_i \otimes \sigma_j)(\sigma_k \otimes \sigma_l)\right] = 4\,\delta_{ik}\delta_{jl} \tag{3.17}$$

ensures that all coefficients in Eq. (3.16) are obtained by direct projection.

Analytical behavior in the reduced phase model. For the phase-evolved Bell state in Eq. (3.12), the correlation tensor is

$$T(\theta) = \begin{pmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & -1 \end{pmatrix}. \tag{3.18}$$

This structure shows that the parameter θ induces a rotation in the transverse x-y correlation plane, while leaving the longitudinal z-component invariant.

The commonly evaluated diagonal correlations are

$$\begin{aligned} \langle \sigma_x \otimes \sigma_x \rangle &= T_{xx} = \cos\theta, \\ \langle \sigma_y \otimes \sigma_y \rangle &= T_{yy} = \cos\theta, \\ \langle \sigma_z \otimes \sigma_z \rangle &= T_{zz} = -1. \end{aligned} \tag{3.19}$$

Interpretation. The reduced phase model modifies only coherence terms, producing a rigid rotation of correlations in the transverse plane. Population correlations remain unchanged, reflecting the purely unitary nature of the transformation.
The correlation tensor is therefore the natural object for distinguishing changes in measurement alignment from actual loss of quantum coherence or entanglement.

Dependence on the local operator basis. The numerical values of the tensor components T_{ij} depend on the choice of local measurement axes used to define the Pauli operators.
Under local unitary transformations,

$$\rho_{AB} \;\rightarrow\; (U_A \otimes U_B)\, \rho_{AB}\, (U_A^\dagger \otimes U_B^\dagger),$$

the correlation tensor transforms according to a corresponding rotation of the local operator basis.
Consequently, individual tensor components are representation-dependent, while physically measurable quantities such as expectation values, entanglement measures, and Bell-inequality violations remain invariant under consistent changes of basis.
This distinction becomes particularly important when introducing arbitrary local measurement axes and optimized CHSH measurements in the following sections.

3.6 Arbitrary Local Measurement Axes

The correlation tensor introduced previously provides a complete description of all two-qubit correlations accessible through local Pauli measurements. This structure enables the prediction of measurement outcomes along arbitrary directions on the Bloch sphere.

A single-qubit observable associated with a unit vector

$$\mathbf{n} = \begin{pmatrix} n_x \\ n_y \\ n_z \end{pmatrix}, \qquad \|\mathbf{n}\| = 1,$$

is defined as

$$\sigma(\mathbf{n}) = \mathbf{n} \cdot \boldsymbol{\sigma} = n_x \sigma_x + n_y \sigma_y + n_z \sigma_z. \tag{3.20}$$

The use of a three-component vector to parametrize measurement directions follows from the structure of single-qubit observables. Any Hermitian operator acting on a two-dimensional Hilbert space can be decomposed as

$$O = \alpha I + \sum_{i \in \{x,y,z\}} n_i \sigma_i, \tag{3.21}$$

where $\alpha \in \mathbb{R}$ and σ_i are the Pauli matrices.

The identity term does not contribute to correlation measurements in the same way as the traceless Pauli part, since it produces only a uniform contribution. As a consequence, the physically relevant dichotomic observables can be represented by the traceless operator $\sigma(\mathbf{n})$.
The condition $\|\mathbf{n}\| = 1$ ensures that $\sigma(\mathbf{n})$ has eigenvalues ± 1, corresponding to a valid projective measurement with two possible outcomes.

For a bipartite state ρ_{AB}, choosing local directions $\mathbf{a}$ and $\mathbf{b}$ leads to the correlation function

$$E(\mathbf{a}, \mathbf{b}) = \mathrm{Tr}\left[\rho_{AB}\left(\sigma(\mathbf{a}) \otimes \sigma(\mathbf{b})\right)\right]. \tag{3.22}$$

Expanding in the Pauli basis gives

$$E(\mathbf{a}, \mathbf{b}) = \sum_{i,j \in \{x,y,z\}} a_i \, T_{ij} \, b_j, \tag{3.23}$$

or, in compact matrix notation,

$$E(\mathbf{a}, \mathbf{b}) = \mathbf{a}^{\mathsf{T}} T \mathbf{b}. \tag{3.24}$$

This expression shows that the correlation tensor completely determines all possible measurement outcomes for local Pauli observables.

Analytical behavior for the reduced phase model. Using Eq. (3.18), the correlation function for the phase-evolved Bell state becomes

$$E(\mathbf{a},\mathbf{b};\theta) = \mathbf{a}^{\mathsf{T}} T(\theta)\mathbf{b}. \tag{3.25}$$

For measurement directions restricted to the equatorial plane,

$$\mathbf{a} = \begin{pmatrix} \cos\alpha \\ \sin\alpha \\ 0 \end{pmatrix}, \qquad \mathbf{b} = \begin{pmatrix} \cos\beta \\ \sin\beta \\ 0 \end{pmatrix},$$

this reduces to

$$E(\alpha,\beta;\theta) = \cos(\theta + \beta - \alpha). \tag{3.26}$$

Interpretation. The phase θ acts as an effective shift of the relative measurement angle. Consequently, phase evolution induced by the fiber is operationally equivalent to a rotation of measurement settings in the transverse plane.

Operational implications. In an experimental setting, arbitrary measurement axes are realized through local unitary transformations, such as wave plates, followed by a fixed computational-basis measurement.
This correspondence allows the theoretical model to reproduce realistic measurement procedures by combining local unitary rotations, standard projective measurements, and correlation evaluation through the tensor structure.

3.7 Bell Inequalities and CHSH Nonlocality

The correlation structure introduced in the previous sections provides a complete operational description of measurement outcomes for arbitrary local observables. This framework makes it possible to formulate quantitative criteria that distinguish correlations compatible with classical physics from those that are intrinsically quantum.

The central question addressed by Bell inequalities is the following: *can the observed correlations between spatially separated subsystems be explained by a classical model based on shared hidden information?*

In a classical description based on local hidden variables, it is assumed that the outcomes of measurements performed on subsystems A and B are determined by a shared underlying variable λ, drawn from a probability distribution $p(\lambda)$. This variable represents all pre-existing information that could influence the measurement results.

For each realization of λ, the outcome of a measurement on subsystem A along direction $\mathbf{a}$ is described by a deterministic response function

$$A(\mathbf{a},\lambda) \in \{-1,+1\},$$

and similarly for subsystem B,

$$B(\mathbf{b}, \lambda) \in \{-1, +1\}.$$

The key physical assumption is locality: the outcome at each subsystem depends only on the local measurement setting and on the shared variable λ, but not on the measurement choice at the distant subsystem.
Since λ is not directly accessible, observable quantities correspond to averages over its distribution. The classical correlation function is therefore [3]

$$E(\mathbf{a}, \mathbf{b}) = \int d\lambda\, p(\lambda)\, A(\mathbf{a}, \lambda)\, B(\mathbf{b}, \lambda), \qquad p(\lambda) \geq 0, \qquad \int d\lambda\, p(\lambda) = 1. \tag{3.27}$$

This expression represents the most general form of bipartite correlations compatible with a local hidden-variable model.

The most widely used form of a Bell inequality is the Clauser–Horne–Shimony–Holt (CHSH) inequality. It involves four measurement settings:

$$\mathbf{a}_1, \mathbf{a}_2 \quad \text{for subsystem } A, \qquad \mathbf{b}_1, \mathbf{b}_2 \quad \text{for subsystem } B.$$

The CHSH parameter is defined as

$$S = E(\mathbf{a}_1, \mathbf{b}_1) + E(\mathbf{a}_1, \mathbf{b}_2) + E(\mathbf{a}_2, \mathbf{b}_1) - E(\mathbf{a}_2, \mathbf{b}_2). \tag{3.28}$$

Classical bound. Under the assumptions of locality and realism, the CHSH parameter satisfies

$$|S| \leq 2. \tag{3.29}$$

This bound reflects a fundamental limitation of classical correlations: although each term in S can individually reach its maximal value, their combination is constrained by the requirement that all outcomes originate from the same underlying variable λ.

Quantum prediction. In quantum mechanics, correlations arise from the structure of the joint state rather than from pre-existing local variables. The correlation function is given by Eq. (3.22).
Due to the non-commuting nature of quantum observables and the presence of entanglement, the CHSH parameter can exceed the classical bound. The maximal value allowed by quantum mechanics is

$$|S| \leq 2\sqrt{2}, \tag{3.30}$$

known as the Tsirelson bound.
This violation does not imply faster-than-light communication. Rather, it implies that the correlations cannot be reproduced by any model based on local hidden variables.

Tensor formulation. Using the correlation tensor representation, the CHSH parameter can be expressed as

$$S = \mathbf{a}_1^{\mathsf{T}} T \mathbf{b}_1 + \mathbf{a}_1^{\mathsf{T}} T \mathbf{b}_2 + \mathbf{a}_2^{\mathsf{T}} T \mathbf{b}_1 - \mathbf{a}_2^{\mathsf{T}} T \mathbf{b}_2. \tag{3.31}$$

This formulation makes explicit that the CHSH inequality depends only on the correlation tensor and on the choice of measurement directions. In particular, it shows that the observed value of S is sensitive not only to the state, but also to the alignment between the measurement axes and the principal directions of the tensor.

A particularly important result is that the maximal violation achievable for a given state can be computed directly from the tensor. Defining

$$U = T^{\mathsf{T}} T, \tag{3.32}$$

and denoting by λ_1 and λ_2 the two largest eigenvalues of U, the maximal CHSH value is

$$S_{\max} = 2\sqrt{\lambda_1 + \lambda_2}. \tag{3.33}$$

Interpretation. A violation of the CHSH inequality, i.e. $|S| > 2$, indicates the presence of correlations that cannot be explained by any local hidden-variable model. Such correlations are referred to as nonlocal.

Nonlocality should be understood as a property of correlations rather than of individual subsystems: each subsystem alone may appear completely random, while the joint statistics reveal strong structured correlations.

It is important to distinguish nonlocality from entanglement. While all maximally entangled pure states achieve maximal quantum violation for suitable measurement axes, not all entangled states violate the CHSH inequality. In particular, sufficiently mixed states may remain within the classical bound despite being entangled.

Operational implications. The CHSH parameter provides a direct bridge between the theoretical description of correlations and experimentally accessible quantities. In practice, it is evaluated by performing measurements along four selected pairs of local axes and combining the resulting correlation values.

The correlation tensor entering this construction can be experimentally reconstructed from a set of bipartite expectation values in the Pauli basis, in a procedure closely related to quantum state tomography. In this sense, the evaluation of CHSH inequalities can be viewed as a specific post-processing of tomographically accessible data.

Two complementary perspectives are particularly relevant:

- evaluation of S for fixed measurement settings, corresponding to a specific experimental configuration,
- computation of $S_{\max}$, which quantifies the maximal nonlocality extractable from the state.

This separation will play a central role in the analysis of fiber-induced effects, where transformations may alter the observed correlations without necessarily reducing the underlying nonlocality.

3.8 From Pure-State Description to Ensemble Representation

The reduced phase model describes a deterministic unitary evolution of a pure entangled state, parametrized by a single phase θ.
In realistic fiber propagation, however, this phase is not necessarily constant. It may fluctuate due to environmental effects such as birefringence variations, mechanical perturbations, and thermal instability. As a consequence, different photon pairs may experience different phase shifts.

When these fluctuations occur on timescales shorter than the measurement resolution, individual realizations cannot be resolved experimentally. The observed system is therefore not described by a single pure state, but by an ensemble of states corresponding to different phase realizations.

In this regime, the appropriate description is given by a density operator

$$\rho = \sum_k p_k |\psi_k\rangle\langle\psi_k|, \tag{3.34}$$

which encodes both classical statistical uncertainty and loss of phase coherence.

In the present context, the effective state is obtained by averaging over the phase distribution:

$$\rho = \mathbb{E}_\theta \left[|\psi(\theta)\rangle\langle\psi(\theta)| \right]. \tag{3.35}$$

This averaging procedure suppresses off-diagonal coherences in the computational basis and leads to a degradation of correlation observables T_{ij}, even though each underlying realization remains a pure state evolving unitarily.

The density operator satisfies

$$\rho^\dagger = \rho, \qquad \rho \geq 0, \qquad \mathrm{Tr}(\rho) = 1, \tag{3.36}$$

ensuring a consistent probabilistic interpretation.

Expectation values are computed as

$$\langle O \rangle = \mathrm{Tr}(\rho O), \tag{3.37}$$

which generalizes the pure-state expression.

Interpretation. The transition from $|\psi(\theta)\rangle$ to ρ reflects a loss of phase information at the level of measurement, rather than a breakdown of the underlying unitary dynamics. The mixed-state description therefore arises from averaging over unresolved degrees of freedom, not necessarily from intrinsic decoherence at the level of individual realizations.

This formalism provides the foundation for modeling phase noise and more general quantum channels, and will be used in Chapter 5 to interpret the behavior of correlation observables under stochastic perturbations.

3.8.1 Bloch-Sphere Interpretation of Mixed States

For a single qubit, any density operator can be represented in Bloch form as

$$\rho = \frac{1}{2}\left(I + \mathbf{r}\cdot\boldsymbol{\sigma}\right), \tag{3.38}$$

where $\mathbf{r} \in \mathbb{R}^3$ is the Bloch vector and

$$\boldsymbol{\sigma} = (\sigma_x, \sigma_y, \sigma_z).$$

Physical states satisfy

$$|\mathbf{r}| \leq 1. \tag{3.39}$$

Pure states correspond to the boundary condition

$$|\mathbf{r}| = 1,$$

and therefore lie on the surface of the Bloch sphere.
Mixed states instead satisfy

$$|\mathbf{r}| < 1,$$

and are represented by Bloch vectors lying inside the Bloch sphere.
The maximally mixed state corresponds to

$$\mathbf{r} = 0, \qquad \rho = \frac{1}{2}I,$$

which corresponds to the center of the Bloch-sphere representation.

Within this geometric representation, unitary evolution acts as a rotation of the Bloch vector and therefore preserves its norm. Consequently, coherent evolution preserves state purity.
By contrast, statistical averaging and depolarizing processes reduce the Bloch-vector magnitude, corresponding to a contraction toward the center of the Bloch ball. This contraction reflects a progressive loss of coherence and polarization information.

This geometric interpretation provides an intuitive visualization of the transition from pure-state unitary dynamics to effective mixed-state evolution under unresolved stochastic perturbations.

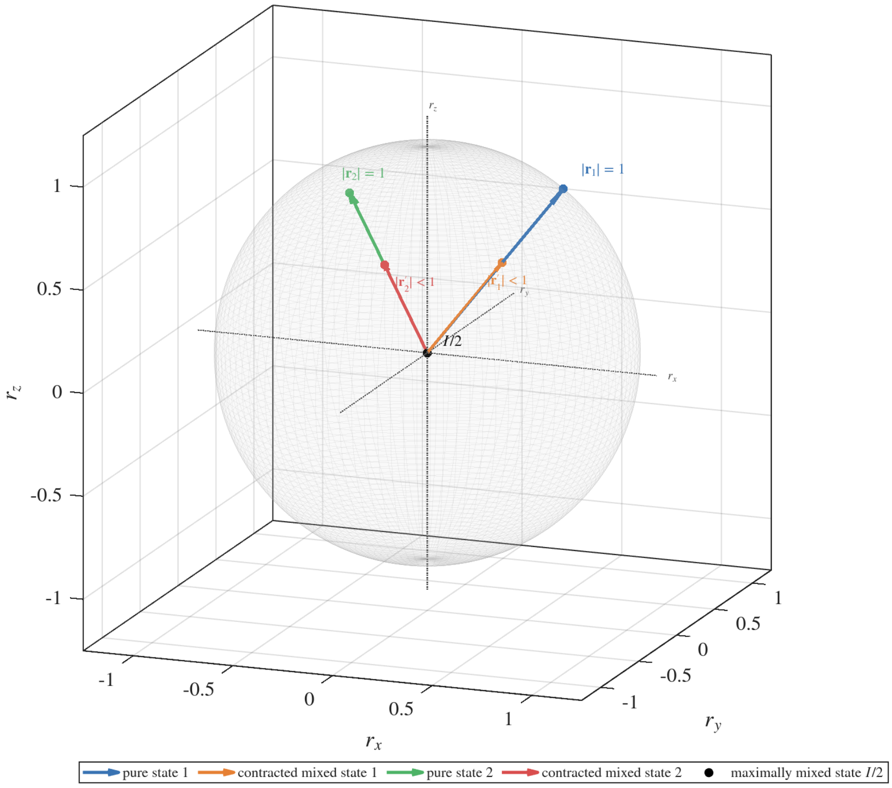


Figure 3.1: Bloch-sphere representation of pure and mixed single-qubit states. Pure states lie on the sphere surface and satisfy $|\mathbf{r}| = 1$, while mixed states are represented by contracted Bloch vectors with $|\mathbf{r}| < 1$.

3.9 Quantum Channels and Depolarization

The ensemble description introduced in the previous section models statistical fluctuations by averaging over a family of unitary realizations. In that framework, each individual evolution remains coherent, and mixedness originates from classical uncertainty over the applied transformations.

More generally, however, a quantum system may interact with additional degrees of freedom that are not observed explicitly. In such cases, the reduced dynamics of the subsystem can no longer be described solely through unitary evolution acting on the state vector. Instead, the evolution must be formulated directly at the level of the density operator.

This motivates the introduction of quantum channels, which provide the general mathematical framework for effective non-unitary evolution.

Quantum channels. A quantum channel is a linear map

$$\mathcal{E} : \rho \mapsto \mathcal{E}(\rho), \tag{3.40}$$

acting on density operators and satisfying the conditions of complete positivity and trace preservation.

A general representation of a quantum channel is given by the operator-sum decomposition

$$\mathcal{E}(\rho) = \sum_{\mu} K_{\mu} \rho K_{\mu}^{\dagger}, \qquad \sum_{\mu} K_{\mu}^{\dagger} K_{\mu} = I, \tag{3.41}$$

where the operators K_{μ} are referred to as Kraus operators and encode the effective action of the physical process.

The channel formalism extends the unitary description introduced previously. While coherent propagation is represented through transformations of the form

$$\rho \to U \rho U^{\dagger},$$

more general physical processes may produce irreversible loss of coherence and therefore require a non-unitary reduced description.

Global depolarizing channel. Among the simplest and most widely used quantum channels is the depolarizing channel, which models isotropic loss of coherence. In the two-qubit setting considered in this work, the global depolarizing model is written as

$$\mathcal{E}_p(\rho) = (1-p)\rho + \frac{p}{4} I_4, \tag{3.42}$$

where I_4 denotes the identity operator on the four-dimensional bipartite Hilbert space and p quantifies the strength of depolarization.

This channel continuously interpolates between coherent evolution and complete mixing, providing a simple benchmark model for the study of entanglement degradation and correlation suppression in bipartite systems.

Application to Bell states. Applying the channel to the Bell state introduced in Eq. (3.5),

$$\rho_0 = |\Psi^+\rangle\langle\Psi^+|,$$

produces the family of mixed states

$$\rho(p) = (1-p)\rho_0 + \frac{p}{4} I_4. \tag{3.43}$$

This construction provides a controlled framework for analyzing the degradation of bipartite quantum correlations under effective noisy evolution.

The global depolarizing model introduced here should be understood as a reference channel rather than as a complete physical model of two-arm fiber propagation. The refinement to independent local depolarizing channels acting on the two propagation arms is developed later in Chapter 5.

3.10 Computational Representation and Implementation Roadmap

The mathematical structures introduced in this chapter define the computational backbone of the simulator. The purpose of the implementation is not to introduce a separate numerical model, but to realize the theoretical framework in a modular and reproducible form.

At MATLAB level, bipartite pure states are represented as 4×1 complex vectors and operators acting on the full two-qubit system are represented as 4×4 complex matrices.
The computational basis ordering is fixed as

$$\{|00\rangle, |01\rangle, |10\rangle, |11\rangle\},$$

and is used consistently across all modules. This convention is critical: tensor products, operator constructions, density matrices, and correlation observables are all defined with respect to this ordering. Any mismatch would lead to incorrect physical results, even if the underlying mathematical expressions are formally correct.

State preparation is implemented through dedicated routines returning normalized Bell states in the fixed computational basis. The state $|\Psi^+\rangle$ is used as the default reference input for the baseline model, providing a controlled benchmark for validating correlation observables, fidelity computations, and entanglement measures under fiber propagation.

The fundamental evolution primitive is the tensor-product action

$$\psi_{\text{out}} = (U_A \otimes U_B)\psi_{\text{in}}.$$

This operation is isolated as a core routine because all unitary propagation models reduce to this structure at the bipartite level. Local single-qubit operators are combined into the global operator $U_A \otimes U_B$, which is then applied to the input state vector.

For the reduced phase model, the simulator is structured around the one-parameter family $|\psi(\theta)\rangle$. Both parametrizations (θ_A, θ_B) and $\theta = \theta_A - \theta_B$ are useful: the two-phase representation preserves the physical interpretation of independent fiber arms, while the reduced one-parameter form enables efficient analytical validation and parameter sweeps.

The first validated numerical study consists of a parameter sweep in θ, with numerical evaluation of state vectors, expectation values, and correlation observables. This defines the minimal complete simulation pipeline upon which all subsequent developments are built.

For correlation analysis, the simulator evaluates tensor components through the density-operator trace rule

$$\langle O \rangle = \text{Tr}(\rho O),$$

using a modular pipeline including:

- Pauli operator generation,

- tensor-product construction,
- expectation-value evaluation,
- structured storage of T_{ij}.

The same computational framework is reused for arbitrary-axis measurements by constructing operators of the form $\sigma(\mathbf{a})\otimes\sigma(\mathbf{b})$, and for CHSH analysis by combining the resulting correlation functions into fixed-axis or optimized Bell parameters.

This organization ensures that changes in the physical model, such as stochastic phase evolution, random unitary propagation, or effective quantum channels, can be introduced by modifying well-isolated components while preserving the validated core routines.

Chapter 4

Quantitative Characterization of Two-Qubit States

4.1 Reduced Density Operators

In bipartite quantum systems, the state of an individual subsystem is described by a density operator obtained through a partial trace over the complementary subsystem. This formalism captures the effect of correlations present at the global level.

Let the composite system be defined on the Hilbert space

$$\mathcal{H}_{AB} = \mathcal{H}_A \otimes \mathcal{H}_B,$$

with $\mathcal{H}_A \cong \mathbb{C}^2$ and $\mathcal{H}_B \cong \mathbb{C}^2$. A generic global state is described by a density operator

$$\rho_{AB} \in \mathbb{C}^{4\times 4}.$$

The reduced density operators are defined as

$$\rho_A = \mathrm{Tr}_B(\rho_{AB}), \qquad \rho_B = \mathrm{Tr}_A(\rho_{AB}). \tag{4.1}$$

These operators reproduce all expectation values of observables acting locally on the subsystem. For any observable O_A acting on subsystem A,

$$\langle O_A \rangle = \mathrm{Tr}[\rho_{AB}\,(O_A \otimes I_B)] = \mathrm{Tr}(\rho_A O_A). \tag{4.2}$$

An explicit construction of the partial trace is given by

$$\rho_A = \sum_{j=0}^{1} (I_A \otimes \langle j|)\rho_{AB}(I_A \otimes |j\rangle), \tag{4.3}$$

which shows that the degrees of freedom associated with the unobserved subsystem are eliminated.

For maximally entangled Bell states, such as

$$|\Psi^+\rangle = \frac{|01\rangle + |10\rangle}{\sqrt{2}},$$

one obtains

$$\rho_A = \rho_B = \frac{I}{2}, \tag{4.4}$$

indicating that each subsystem is locally maximally mixed despite the global state being pure.

The same result holds for the phase-evolved state introduced in Chapter 3,

$$|\psi(\theta)\rangle = \frac{|01\rangle + e^{i\theta}|10\rangle}{\sqrt{2}},$$

since the relative phase modifies global coherence but leaves local states unchanged:

$$\rho_A(\theta) = \rho_B(\theta) = \frac{I}{2}. \tag{4.5}$$

By linearity of the partial trace, this property is preserved under convex combinations. For the ensemble state

$$\rho_{AB}^{(\text{ens})} = \frac{1}{N}\sum_{k=1}^{N} |\psi(\theta_k)\rangle\langle\psi(\theta_k)|,$$

one finds

$$\rho_A^{(\text{ens})} = \rho_B^{(\text{ens})} = \frac{I}{2}. \tag{4.6}$$

This leads to a key conceptual point: the same reduced density operator

$$\rho_A = \frac{I}{2}$$

can arise from physically distinct global states.

On one hand, it may describe a classical ensemble

$$\rho^{(\text{class})} = \frac{1}{2}|0\rangle\langle 0| + \frac{1}{2}|1\rangle\langle 1|,$$

where the system is in a definite state but the observer lacks knowledge.
On the other hand, it may result from entanglement:

$$\rho_A^{(\text{ent})} = \mathrm{Tr}_B\left(|\Psi^+\rangle\langle\Psi^+|\right) = \frac{I}{2}.$$

Since both cases yield identical expectation values for all local observables,

$$\mathrm{Tr}(\rho^{(\text{class})}O_A) = \mathrm{Tr}(\rho_A^{(\text{ent})}O_A),$$

they are indistinguishable by local measurements alone.
The distinction emerges only at the global level, where coherence and correlations must be analyzed through quantities such as purity and bipartite correlation functions.

Finally, the reduced density operator directly determines measurable local quantities. For Pauli observables,

$$\begin{aligned}\langle\sigma_x\rangle &= \mathrm{Tr}(\rho_A\sigma_x),\\ \langle\sigma_y\rangle &= \mathrm{Tr}(\rho_A\sigma_y),\\ \langle\sigma_z\rangle &= \mathrm{Tr}(\rho_A\sigma_z).\end{aligned} \tag{4.7}$$

For $\rho_A = I/2$, all expectation values vanish, indicating complete local depolarization.

4.2 Purity and Mixedness

Having introduced reduced density operators in Section 4.1, we now require a quantitative criterion to distinguish whether a quantum state is pure or mixed.

A state described by a density operator ρ is pure if it can be written as

$$\rho = |\psi\rangle\langle\psi|, \tag{4.8}$$

for some normalized vector $|\psi\rangle$. Otherwise, it is mixed, meaning that it admits an ensemble representation

$$\rho = \sum_k p_k|\psi_k\rangle\langle\psi_k|, \qquad p_k \geq 0, \qquad \sum_k p_k = 1. \tag{4.9}$$

A convenient quantitative measure of mixedness is the purity, defined as

$$\gamma(\rho) = \mathrm{Tr}(\rho^2). \tag{4.10}$$

This quantity satisfies

$$\gamma(\rho) = 1 \iff \rho \text{ is pure}, \qquad \gamma(\rho) < 1 \iff \rho \text{ is mixed}. \tag{4.11}$$

If $\rho = \sum_i \lambda_i|i\rangle\langle i|$ is its spectral decomposition, then

$$\gamma(\rho) = \sum_i \lambda_i^2, \tag{4.12}$$

which shows that purity measures the concentration of the eigenvalue distribution. More generally, for a Hilbert space of dimension d,

$$\frac{1}{d} \leq \gamma(\rho) \leq 1, \tag{4.13}$$

with the lower bound attained by the maximally mixed state $\rho = I/d$.

For single-qubit systems, purity admits a simple geometric interpretation. Any density operator can be written as

$$\rho = \frac{1}{2}\left(I + \vec{r}\cdot\vec{\sigma}\right), \tag{4.14}$$

where $\vec{r} \in \mathbb{R}^3$ is the Bloch vector. One then finds

$$\gamma(\rho) = \frac{1 + \|\vec{r}\|^2}{2}. \tag{4.15}$$

Thus,

$$\|\vec{r}\| = 1 \Rightarrow \rho \text{ is pure}, \qquad \|\vec{r}\| = 0 \Rightarrow \rho = \frac{I}{2}.$$

Mixedness does not uniquely determine the physical origin of a state. As discussed in Section 4.1, it may arise either from classical statistical uncertainty or from entanglement with an unobserved subsystem.

A paradigmatic example is provided by the Bell state

$$|\Psi^+\rangle = \frac{|01\rangle + |10\rangle}{\sqrt{2}},$$

whose global density operator

$$\rho_{AB} = |\Psi^+\rangle\langle\Psi^+|$$

is pure,

$$\gamma(\rho_{AB}) = 1,$$

while the reduced states satisfy

$$\rho_A = \rho_B = \frac{I}{2}, \qquad \gamma(\rho_A) = \gamma(\rho_B) = \frac{1}{2}.$$

This illustrates that a globally pure entangled state may generate locally maximally mixed subsystems.

In the present model, the phase-evolved state introduced in Chapter 3,

$$|\psi(\theta)\rangle = \frac{|01\rangle + e^{i\theta}|10\rangle}{\sqrt{2}},$$

remains pure for all values of θ,

$$\gamma(\rho(\theta)) = 1, \tag{4.16}$$

while the reduced states remain maximally mixed,

$$\rho_A(\theta) = \rho_B(\theta) = \frac{I}{2}.$$

Thus, relative phase shifts modify correlation observables without affecting either global purity or local mixedness.

When extending the model to include stochastic or environmental effects, the situation changes. For the random-phase ensemble, the global state depends on the coherence parameter

$$\mu = \langle e^{i\theta}\rangle, \tag{4.17}$$

and its purity is

$$\gamma\left(\rho_{AB}^{(\mathrm{ens})}\right) = \frac{1 + |\mu|^2}{2}. \tag{4.18}$$

Equivalently,

$$\gamma\left(\rho_{AB}^{(\mathrm{ens})}\right) = \frac{1 + \langle\cos\theta\rangle^2 + \langle\sin\theta\rangle^2}{2}. \tag{4.19}$$

This shows that purity directly quantifies the loss of phase coherence in the ensemble: the state is pure for $|\mu| = 1$ and becomes mixed as $|\mu|$ decreases.

Purity therefore provides a fundamental diagnostic tool for assessing coherence loss in the simulator, complementing correlation observables and preparing the ground for fidelity and entanglement measures.

4.3 Fidelity with Respect to Bell States

Having introduced reduced density operators and purity as measures of local information and mixedness, we now require a quantitative tool to evaluate how close a given quantum state is to a reference target state. This role is fulfilled by the notion of fidelity.

In general, the fidelity between two quantum states described by density operators ρ and σ is defined as

$$F(\rho, \sigma) = \left(\mathrm{Tr}\sqrt{\sqrt{\rho}\,\sigma\,\sqrt{\rho}}\right)^2, \tag{4.20}$$

and satisfies

$$0 \leq F(\rho, \sigma) \leq 1,$$

with equality to 1 if and only if $\rho = \sigma$.

In the present context, the reference state is pure, $\sigma = |\phi\rangle\langle\phi|$, and the expression simplifies to

$$F(\rho, |\phi\rangle) = \langle\phi|\rho|\phi\rangle, \tag{4.21}$$

which admits a direct interpretation as the probability of projecting onto $|\phi\rangle$.

We focus on the Bell state introduced in Chapter 3,

$$|\Psi^+\rangle = \frac{|01\rangle + |10\rangle}{\sqrt{2}},$$

and define

$$F_{\Psi^+}(\rho) = \langle\Psi^+|\rho|\Psi^+\rangle. \tag{4.22}$$

For the phase-evolved state,

$$|\psi(\theta)\rangle = \frac{|01\rangle + e^{i\theta}|10\rangle}{\sqrt{2}}, \qquad \rho(\theta) = |\psi(\theta)\rangle\langle\psi(\theta)|,$$

one obtains

$$F_{\Psi^+}(\rho(\theta)) = |\langle\Psi^+|\psi(\theta)\rangle|^2,$$

with

$$\langle \Psi^+ | \psi(\theta) \rangle = \frac{1}{2}\left(1 + e^{i\theta}\right),$$

leading to

$$F_{\Psi^+}(\rho(\theta)) = \frac{1 + \cos\theta}{2}. \tag{4.23}$$

Thus, the relative phase modifies the overlap with the reference Bell state, even though the global state remains pure. In particular,

$$F_{\Psi^+} = 1 \quad \text{for } \theta = 0, \qquad F_{\Psi^+} = 0 \quad \text{for } \theta = \pi.$$

For ensemble states,

$$\rho_{AB}^{(\mathrm{ens})} = \frac{1}{N} \sum_{k=1}^{N} |\psi(\theta_k)\rangle\langle\psi(\theta_k)|,$$

linearity yields

$$F_{\Psi^+}\left(\rho_{AB}^{(\mathrm{ens})}\right) = \frac{1}{N} \sum_{k=1}^{N} F_{\Psi^+}\left(\rho(\theta_k)\right).$$

Introducing the coherence parameter $\mu = \langle e^{i\theta} \rangle$, one finds

$$F_{\Psi^+} = \frac{1}{2}\left(1 + \mathrm{Re}(\mu)\right) = \frac{1}{2}\left(1 + \langle \cos\theta \rangle\right). \tag{4.24}$$

In particular, for a uniform distribution over $[0, 2\pi]$,

$$F_{\Psi^+} = \frac{1}{2}.$$

Fidelity therefore provides a quantitative measure of how closely a given state approximates the target Bell state. Unlike purity, which characterizes mixedness, fidelity is sensitive to coherent deformations of the state and complements correlation observables in the analysis of the system.

4.4 Concurrence as an Entanglement Measure

Reduced density operators, purity, and fidelity provide complementary information about a bipartite quantum state, but none of them directly quantifies entanglement. In particular, fidelity with respect to a Bell state may vary under local unitary evolution even when the amount of entanglement remains unchanged. A genuine entanglement measure is therefore required.

For two-qubit systems, a standard measure is the concurrence, defined such that

$$0 \leq C \leq 1,$$

where $C = 0$ corresponds to separable states and $C = 1$ to maximally entangled states.

For a pure state

$$|\psi\rangle = a|00\rangle + b|01\rangle + c|10\rangle + d|11\rangle,$$

the concurrence is

$$C(|\psi\rangle) = 2|ad - bc|. \tag{4.25}$$

This expression vanishes for separable states and reaches unity for Bell states. For example,

$$|\Psi^+\rangle = \frac{|01\rangle + |10\rangle}{\sqrt{2}} \quad \Rightarrow \quad C = 1.$$

For the phase-evolved state introduced in Chapter 3,

$$|\psi(\theta)\rangle = \frac{|01\rangle + e^{i\theta}|10\rangle}{\sqrt{2}},$$

one finds

$$C(|\psi(\theta)\rangle) = 1, \tag{4.26}$$

showing that the relative phase modifies fidelity but leaves entanglement unchanged. This reflects the invariance of bipartite entanglement under local unitary transformations.

For pure bipartite states, concurrence is directly related to the reduced density operators:

$$C(|\psi\rangle) = 2\sqrt{\det(\rho_A)} = \sqrt{2\,(1 - \gamma(\rho_A))}, \qquad \rho_A = \mathrm{Tr}_B\left(|\psi\rangle\langle\psi|\right). \tag{4.27}$$

Thus, entanglement is encoded in the mixedness of the reduced states: separable states yield pure reductions, while maximally entangled states yield maximally mixed reductions.

For mixed states $\rho \in \mathbb{C}^{4\times 4}$, concurrence is defined via the spin-flipped matrix

$$\tilde{\rho} = (\sigma_y \otimes \sigma_y)\rho^*(\sigma_y \otimes \sigma_y). \tag{4.28}$$

Let $\lambda_1 \geq \lambda_2 \geq \lambda_3 \geq \lambda_4 \geq 0$ be the square roots of the eigenvalues of $\rho\tilde{\rho}$. Then

$$C(\rho) = \max\left(0, \lambda_1 - \lambda_2 - \lambda_3 - \lambda_4\right). \tag{4.29}$$

This expression applies to arbitrary two-qubit states and reduces to the pure-state formula in the appropriate limit.

In the present model, mixed states arise from ensemble averaging over random phases,

$$\rho_{AB}^{(\mathrm{ens})} = \frac{1}{N}\sum_{k=1}^{N} |\psi(\theta_k)\rangle\langle\psi(\theta_k)|,$$

which depend on the coherence parameter

$$\mu = \langle e^{i\theta} \rangle.$$

In the computational basis, the ensemble state reads

$$\rho_{AB}^{(\mathrm{ens})} = \frac{1}{2}|01\rangle\langle 01| + \frac{1}{2}|10\rangle\langle 10| + \frac{\mu}{2}|10\rangle\langle 01| + \frac{\mu^*}{2}|01\rangle\langle 10|. \tag{4.30}$$

For this family of states, the concurrence is

$$C\left(\rho_{AB}^{(\mathrm{ens})}\right) = |\mu|. \tag{4.31}$$

This result shows that the same coherence parameter controlling purity and fidelity also governs entanglement degradation:

$$\gamma\left(\rho_{AB}^{(\mathrm{ens})}\right) = \frac{1+|\mu|^2}{2}, \qquad F_{\Psi^+} = \frac{1+\mathrm{Re}(\mu)}{2}.$$

Thus, the reduced phase model provides a unified description in which coherence, mixedness, and entanglement are all controlled by a single parameter.

4.5 Interplay Between Purity, Fidelity, and Concurrence

The previous sections introduced three complementary descriptors of bipartite quantum states: purity, fidelity, and concurrence. While purity quantifies mixedness, fidelity measures proximity to a chosen reference state, and concurrence provides a direct measure of bipartite entanglement.

In general, these quantities are independent. However, within the random-phase ensemble model considered in this work, they are all governed by a single complex parameter, the ensemble coherence

$$\mu = \langle e^{i\theta} \rangle = \langle \cos\theta \rangle + i\langle \sin\theta \rangle. \tag{4.32}$$

For this family of states, the three quantities take the form

$$\gamma(\rho) = \frac{1+|\mu|^2}{2}, \qquad C(\rho) = |\mu|, \qquad F_{\Psi^+} = \frac{1+\mathrm{Re}(\mu)}{2}. \tag{4.33}$$

These relations highlight the distinct roles of the three quantities. Purity depends on the squared magnitude $|\mu|^2$ and therefore quantifies the overall coherence retained after ensemble averaging. Concurrence depends linearly on $|\mu|$ and directly measures entanglement. By contrast, fidelity depends only on the real part $\mathrm{Re}(\mu)$, reflecting overlap with a specific reference state.

This distinction has important consequences. In the deterministic phase model, the parameter μ lies on the unit circle, so that $|\mu| = 1$ for all values of θ. As a result, concurrence remains maximal, while fidelity varies with the phase. Thus, fidelity changes do not necessarily indicate variations in entanglement.

In the random-phase ensemble, averaging reduces the magnitude $|\mu|$. This leads to a simultaneous decrease of purity and concurrence, reflecting a genuine loss of coherence and entanglement. In this regime, fidelity reduction is no longer purely geometric but corresponds to physical degradation of the state.

A useful geometric interpretation is obtained by representing μ in the complex plane. The concurrence is proportional to its distance from the origin, while fidelity corresponds to its projection onto the real axis. Deterministic phase evolution moves μ along the unit circle, preserving its magnitude, whereas ensemble averaging contracts it toward the origin.

The limiting cases illustrate these behaviors:

$$|\mu| = 1 \Rightarrow \gamma(\rho) = 1, \quad C = 1, \qquad |\mu| = 0 \Rightarrow \gamma(\rho) = \frac{1}{2}, \quad C = 0. \tag{4.34}$$

Intermediate values $0 < |\mu| < 1$ correspond to partially coherent states with reduced but nonzero entanglement.

These relations are specific to the structure of the random-phase ensemble. In general quantum states, purity, fidelity, and concurrence are not determined by a single parameter.

In summary, the coherence parameter μ provides a unifying description of the states considered in this work: its magnitude controls both mixedness and entanglement, while its real part determines the overlap with the reference Bell state. This interplay highlights the necessity of combining multiple quantitative descriptors to fully characterize bipartite quantum systems.

4.6 Computational Evaluation of State Metrics

The quantities introduced in this chapter define the main diagnostic layer of the simulator. Their role is to translate the abstract density-operator description into numerical indicators of local mixedness, global coherence, Bell-state overlap, and entanglement.

At the computational level, reduced density operators are obtained by applying partial-trace routines to the global density matrix ρ_{AB}. For a two-qubit state represented in the fixed computational basis, this operation returns 2×2 matrices ρ_A and ρ_B, which can then be used to compute local expectation values and local purities.

Purity is evaluated through the trace formula

$$\gamma(\rho) = \mathrm{Tr}(\rho^2),$$

and is applied both to the global state and to the reduced states. This provides a direct numerical check of whether a simulated state remains pure, becomes mixed through ensemble averaging, or approaches the maximally mixed limit.

Fidelity with respect to $|\Psi^+\rangle$ is computed using the pure-reference formula

$$F_{\Psi^+}(\rho) = \langle \Psi^+ | \rho | \Psi^+ \rangle,$$

which avoids unnecessary matrix square-root operations and is therefore the natural implementation for the present simulator.

Concurrence is computed using the two-qubit spin-flip construction. For pure states, the simplified formula $C(|\psi\rangle) = 2|ad - bc|$ provides a useful analytical validation. For density matrices, the general mixed-state formula based on the eigenvalues of $\rho\tilde{\rho}$ is used.

The same metric routines are applied across deterministic phase evolution, random-phase ensembles, and depolarizing-channel simulations. This ensures that all numerical experiments are characterized using a common diagnostic framework.

The computational objective is therefore not only to evaluate individual metrics, but also to compare them consistently. In particular:

- purity diagnoses mixedness and coherence loss,
- fidelity measures proximity to the chosen Bell reference state,
- concurrence quantifies bipartite entanglement.

This separation is essential for the interpretation of fiber-induced effects. A decrease in fidelity may simply indicate a coherent phase rotation, while a decrease in concurrence indicates genuine entanglement degradation. Purity distinguishes whether the state remains pure or has become mixed due to unresolved statistical or environmental degrees of freedom.

Chapter 5

Effective Fiber Models

5.1 Motivation for Effective Fiber Models

The previous chapters established the mathematical framework required to describe bipartite polarization states, local unitary propagation, correlation measurements, quantum channels, and quantitative state metrics. Within that framework, fiber propagation was initially modeled through analytically controlled transformations, including deterministic phase shifts, statistical phase ensembles, and idealized depolarizing maps.

These models provide a necessary validation baseline, but they do not yet fully capture the physical structure of a fiber-based quantum communication link. In the experimental scenario considered in this work, the two photons propagate through two distinct optical arms. Each arm acts locally on the polarization degree of freedom of the corresponding photon and may introduce both coherent and incoherent perturbations. Consequently, a physically meaningful fiber model must preserve the bipartite structure of the system while allowing independent evolution and degradation mechanisms on subsystems A and B.

The objective of the present chapter is therefore to construct effective fiber models connecting the abstract quantum-channel formalism introduced previously with the physical interpretation of entangled-photon propagation in optical fibers. The term "effective" is used deliberately: the goal is not to reproduce the full electromagnetic propagation problem inside the fiber, including all microscopic temporal and spectral degrees of freedom, but rather to develop reduced models describing the observable action of the fiber on the polarization state.

From coherent propagation to effective noisy evolution. In the ideal coherent regime, each fiber arm is represented by a local unitary transformation acting on the corresponding polarization qubit. The bipartite state therefore evolves according to

$$\rho_{AB}^{\mathrm{out}} = (U_A \otimes U_B)\, \rho_{AB}^{\mathrm{in}} \left(U_A^\dagger \otimes U_B^\dagger\right). \tag{5.1}$$

In this limit, the evolution preserves purity and entanglement, and the fiber acts as a coherent optical element producing deterministic polarization rotations and relative phase shifts.

Real optical fibers, however, may introduce additional propagation effects that cannot be represented solely through fixed polarization rotations. Unresolved temporal, spectral, and environmental fluctuations may effectively reduce polarization coherence when only polarization observables are retained in the description. At the level of the reduced polarization state, these effects appear as non-unitary dynamics.

Within the density-operator formalism introduced previously, such mechanisms are described through effective local quantum channels acting independently on the two subsystems:

$$\rho_{AB}^{\text{out}} = (\mathcal{E}_A \otimes \mathcal{E}_B)\left(\rho_{AB}^{\text{in}}\right), \tag{5.2}$$

where $\mathcal{E}_A$ and $\mathcal{E}_B$ describe the effective action of the two fiber arms.

Entanglement sensitivity to local degradation. Entanglement is encoded in the nonlocal correlations of the joint bipartite state ρ_{AB}, rather than in the properties of either subsystem separately. Consequently, even moderate local perturbations acting on a single propagation arm may significantly degrade global quantum correlations.

This aspect is particularly relevant for quantum communication systems because many protocols rely directly on the preservation of bipartite entanglement. Bell-state fidelity, concurrence, correlation tensors, and CHSH nonlocality may all degrade even when local polarization statistics remain apparently reasonable.

For this reason, the physically relevant object is the full bipartite density operator together with its associated correlation and entanglement structure, rather than the reduced states ρ_A and ρ_B alone.

Modeling strategy of the present chapter. The construction developed in this chapter proceeds progressively. First, local depolarizing channels are introduced as effective models of independent polarization degradation in the two fiber arms. These channels are then combined with the coherent phase-evolution model developed previously, leading to composite effective fiber channels incorporating both residual unitary evolution and incoherent depolarization.

This progression establishes the bridge between the analytically controlled models introduced previously and the more physically meaningful effective propagation models studied later in the numerical experiments.

5.2 Local Depolarizing Fiber Channels

The global depolarizing model introduced previously provides a useful baseline for studying the degradation of bipartite quantum correlations. However, that model acts collectively on the full two-qubit state and does not explicitly distinguish the physical propagation of the two photons through separate optical fibers.

In realistic fiber-based quantum communication systems, the two photons propagate through distinct optical paths. As a consequence, polarization degradation may occur independently in the two propagation arms. This motivates the introduction of local depolarizing channels acting separately on subsystems A and B.

Effective local channel model. The effective evolution is modeled through independent local channels acting on the two subsystems:

$$\rho_{AB}^{\text{out}} = (\mathcal{D}_{p_A} \otimes \mathcal{D}_{p_B}) \left(\rho_{AB}^{\text{in}}\right), \tag{5.3}$$

where $\mathcal{D}_{p_A}$ and $\mathcal{D}_{p_B}$ denote single-qubit depolarizing channels characterized by independent depolarization parameters p_A and p_B.

It is important to emphasize that the local channels act on the full bipartite density operator ρ_{AB}, rather than independently on the reduced states ρ_A and ρ_B.

Although the physical propagation occurs locally in each arm, the relevant quantum correlations are encoded in the nonlocal structure of the joint density operator. In particular, entanglement and bipartite correlation tensors cannot, in general, be reconstructed from the reduced density operators alone.

For maximally entangled Bell states, for example, the reduced states remain maximally mixed,

$$\rho_A = \rho_B = \frac{I}{2},$$

independently of the entanglement structure of the global state. Consequently, evolving only the reduced density operators would fail to capture the degradation of bipartite quantum correlations induced by local propagation noise.

The tensor-product channel structure in Eq. (5.3) therefore preserves the full operator structure of the bipartite state while still representing physically local propagation effects.

For this reason, the action of a local channel on the bipartite state must be understood as an operator-map action on one tensor factor, not as an ordinary matrix multiplication. If a bipartite operator is written as a linear combination of product operators,

$$X_{AB} = \sum_k A_k \otimes B_k,$$

then a channel acting locally on subsystem A is applied as

$$(\mathcal{D}_{p_A} \otimes \mathcal{I}) (X_{AB}) = \sum_k \mathcal{D}_{p_A}(A_k) \otimes B_k. \tag{5.4}$$

This representation is always possible in finite-dimensional bipartite systems, since product operators span the composite operator space.

Using the single-qubit operator-map form of the depolarizing channel,

$$\mathcal{D}_p(X) = (1-p)X + p\frac{I_2}{2}\operatorname{Tr}(X). \tag{5.5}$$

one obtains the single-arm action

$$(\mathcal{D}_{p_A} \otimes \mathcal{I}) (\rho_{AB}) = (1-p_A)\rho_{AB} + p_A \left(\frac{I_2}{2} \otimes \rho_B\right), \tag{5.6}$$

where

$$\rho_B = \mathrm{Tr}_A(\rho_{AB}).$$

Analogously,

$$(\mathcal{I} \otimes \mathcal{D}_{p_B})\,(\rho_{AB}) = (1-p_B)\rho_{AB} + p_B\left(\rho_A \otimes \frac{I_2}{2}\right), \tag{5.7}$$

with

$$\rho_A = \mathrm{Tr}_B(\rho_{AB}).$$

Applying the two single-arm channels in sequence gives

$$\begin{aligned}
(\mathcal{D}_{p_A} \otimes \mathcal{D}_{p_B})\,(\rho_{AB}) = {} & (1-p_A)(1-p_B)\rho_{AB} \\
& + p_A(1-p_B)\left(\frac{I_2}{2} \otimes \rho_B\right) \\
& + (1-p_A)p_B\left(\rho_A \otimes \frac{I_2}{2}\right) \\
& + p_A p_B\left(\frac{I_2}{2} \otimes \frac{I_2}{2}\right).
\end{aligned} \tag{5.8}$$

This expression makes explicit that local depolarization is local in its physical action, but it is still applied to the full joint state. The reduced density operators appear only because complete depolarization of one subsystem replaces that subsystem with $I_2/2$ while preserving the marginal state of the other subsystem.

This type of bipartite local channel has been studied extensively in the theory of composite quantum channels, including the characterization of positivity, complete positivity, entanglement-breaking, and entanglement-annihilating regimes for tensor products of depolarizing maps [4].

Single-qubit depolarizing action. For a single-qubit density operator represented in Bloch form as

$$\rho = \frac{1}{2}\left(I + \mathbf{r}\cdot\boldsymbol{\sigma}\right),$$

the depolarizing channel is modeled through

$$\mathcal{D}_p(\rho) = \frac{1}{2}\left(I + (1-p)\,\mathbf{r}\cdot\boldsymbol{\sigma}\right). \tag{5.9}$$

As discussed previously, the depolarizing channel acts as a linear operator map on the space of density operators rather than as an ordinary matrix multiplication. In the Bloch representation, this action corresponds to an isotropic contraction of the polarization vector.

Therefore, the depolarizing process cannot be interpreted simply as multiplication of the density operator by a scalar matrix factor. Instead, the channel replaces part of the operator structure by the maximally mixed contribution while preserving trace normalization.

This operator-map structure becomes particularly important in the bipartite setting, where local depolarization acts on the full joint density operator and modifies the correlation structure of the composite state.

The action of the channel therefore corresponds to an isotropic contraction of the Bloch vector:

$$\mathbf{r} \to (1-p)\mathbf{r}.$$

Consequently:

- $p = 0$ corresponds to ideal coherent propagation,
- $0 < p < 1$ describes partial polarization degradation,
- $p = 1$ maps the state to the maximally mixed state $I/2$.

This representation provides a simple and analytically tractable effective description of polarization decoherence while preserving a direct geometric interpretation at the level of the Bloch sphere.

Difference from global depolarization. It is important to distinguish local two-arm depolarization from the global depolarizing model introduced previously.

In the global model, the depolarizing map acts collectively on the full Hilbert space. For a d-dimensional system, the corresponding operator-map form is

$$\mathcal{D}_p^{(d)}(X) = (1-p)X + p\frac{I_d}{d}\,\mathrm{Tr}(X). \tag{5.10}$$

For a normalized density operator ρ, this reduces to

$$\mathcal{D}_p^{(d)}(\rho) = (1-p)\rho + p\frac{I_d}{d}. \tag{5.11}$$

In the two-qubit case considered here, $d = 4$, so that

$$\rho \to (1-p)\rho + \frac{p}{4}I_4. \tag{5.12}$$

Here the depolarization acts collectively on the full bipartite system.

In contrast, the local model acts independently on the two physical propagation arms through the tensor-product channel

$$\mathcal{D}_{p_A} \otimes \mathcal{D}_{p_B}.$$

Although both models produce suppression of quantum correlations and progressive entanglement degradation, they correspond to physically distinct propagation mechanisms.

The local model is therefore more naturally aligned with the physical structure of fiber-based bipartite propagation, where each photon experiences an independent effective environment during transmission.

Role in the modeling hierarchy. The local depolarizing model constitutes the first effective fiber-channel model in which independent incoherent degradation mechanisms are explicitly associated with the two propagation arms.

This construction prepares the introduction of more advanced effective models combining coherent phase evolution and local depolarization within a unified composite propagation channel.

5.3 Two-Arm Correlation Tensor Contraction

The local depolarizing model introduced in the previous section has a direct and experimentally meaningful effect on the bipartite correlation structure of the propagated state. Since polarization correlations are represented through expectation values of Pauli-product observables, the correlation tensor provides the natural framework for describing how local fiber degradation modifies measurable quantities.

The effective two-arm depolarizing channel is

$$\mathcal{E}_{p_A,p_B} = \mathcal{D}_{p_A} \otimes \mathcal{D}_{p_B}, \tag{5.13}$$

where $\mathcal{D}_{p_A}$ and $\mathcal{D}_{p_B}$ act independently on subsystems A and B.

The single-qubit depolarizing channel preserves the identity operator and contracts all traceless Pauli components isotropically:

$$\mathcal{D}_p(I) = I, \qquad \mathcal{D}_p(\sigma_i) = (1-p)\sigma_i, \qquad i \in \{x, y, z\}. \tag{5.14}$$

Consequently, for a generic two-body Pauli product $\sigma_i \otimes \sigma_j$, one obtains

$$\begin{aligned} \mathcal{E}_{p_A,p_B}(\sigma_i \otimes \sigma_j) &= \mathcal{D}_{p_A}(\sigma_i) \otimes \mathcal{D}_{p_B}(\sigma_j) \\ &= (1-p_A)(1-p_B)\sigma_i \otimes \sigma_j. \end{aligned} \tag{5.15}$$

It follows that all bipartite Pauli correlations are contracted by the common factor

$$\eta = (1-p_A)(1-p_B). \tag{5.16}$$

For a generic two-qubit state, the correlation tensor components are defined by

$$T_{ij} = \mathrm{Tr}\left(\rho_{AB}\sigma_i \otimes \sigma_j\right), \qquad i, j \in \{x, y, z\}. \tag{5.17}$$

Using Eq. (5.15), the output tensor satisfies

$$T_{ij}^{\mathrm{out}} = \eta T_{ij}^{\mathrm{in}}, \qquad i, j \in \{x, y, z\}. \tag{5.18}$$

Equivalently, in matrix form,

$$T_{\mathrm{out}} = \eta T_{\mathrm{in}}. \tag{5.19}$$

The quantity η therefore represents the effective survival factor of bipartite polarization correlations under independent local depolarization in the two propagation arms.

Physical interpretation. The tensor-contraction law derived above provides a simple geometric interpretation of isotropic local depolarization. The local channels do not alter the structural form of the correlation tensor; instead, they uniformly reduce its magnitude.

Physically, this means that local depolarization suppresses the strength of polarization correlations without introducing preferred directions in polarization space. Within the isotropic approximation adopted here, all two-body correlations are degraded uniformly.

This behavior differs conceptually from coherent phase evolution. A local phase shift modifies the orientation of the correlation structure relative to a fixed measurement basis, while isotropic depolarization reduces the overall correlation amplitude.

Operational consequences. For arbitrary local measurement directions $\mathbf{a}$ and $\mathbf{b}$, the correlation function is

$$E(\mathbf{a}, \mathbf{b}) = \mathbf{a}^T T \mathbf{b}.$$

Using Eq. (5.19), the corresponding output correlation function becomes

$$E_{\text{out}}(\mathbf{a}, \mathbf{b}) = \eta E_{\text{in}}(\mathbf{a}, \mathbf{b}). \tag{5.20}$$

Therefore, within the isotropic local-depolarization model, every two-arm polarization correlation is suppressed by the same multiplicative factor η.

Since Bell-inequality parameters such as the CHSH quantity are constructed from linear combinations of these correlation functions, the same contraction factor directly affects the corresponding nonlocality indicators.

The tensor-contraction law established in this section constitutes the key analytical ingredient for the composite phase–depolarization fiber model developed in the following section.

5.4 Composite Phase–Depolarization Fiber Channel

The models introduced in the previous sections describe two complementary physical mechanisms affecting polarization-entangled photon propagation in optical fibers.

On one hand, coherent phase evolution modifies the orientation of the bipartite correlation structure through local unitary transformations. On the other hand, local depolarizing channels reduce the magnitude of polarization correlations through effective non-unitary dynamics.

A realistic reduced fiber model should therefore combine both effects simultaneously. This motivates the introduction of a composite effective channel in which coherent phase evolution and local depolarization act together on the bipartite polarization state.

5.4.1 Definition of the Composite Channel

The effective evolution is modeled through the composition

$$\rho_{AB}^{\mathrm{out}} = (\mathcal{D}_{p_A} \otimes \mathcal{D}_{p_B}) \left[U_\theta \rho_{AB}^{\mathrm{in}} U_\theta^\dagger \right], \tag{5.21}$$

where

$$U_\theta = E_\theta \otimes I, \tag{5.22}$$

with

$$E_\theta = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\theta} \end{pmatrix}. \tag{5.23}$$

The parameter θ represents the residual coherent phase shift accumulated during propagation, while the depolarization parameters p_A and p_B quantify the effective incoherent polarization degradation occurring independently in the two fiber arms.

As demonstrated in Appendix C.2, isotropic local depolarization commutes with the reduced phase-evolution model considered here. Explicitly,

$$(\mathcal{D}_{p_A} \otimes \mathcal{D}_{p_B}) \left[U_\theta \rho U_\theta^\dagger \right] = U_\theta \left[(\mathcal{D}_{p_A} \otimes \mathcal{D}_{p_B}) (\rho) \right] U_\theta^\dagger. \tag{5.24}$$

Therefore, within the isotropic approximation adopted in this work, the ordering of the coherent and incoherent contributions is not relevant.

This commutation property is not expected to hold in general for arbitrary noise models. In particular, anisotropic polarization degradation, axis-dependent decoherence mechanisms, or dynamically varying local fiber transformations may break the commutation relation between coherent evolution and non-unitary dynamics.

In such cases, the ordering of successive channel contributions becomes physically relevant, and the propagation may require a segmented or infinitesimal description in which coherent and incoherent effects are applied sequentially along the fiber evolution.

The isotropic approximation adopted here therefore represents a simplified effective regime in which the analytical structure of the model remains tractable while preserving the essential interplay between coherent phase evolution and bipartite correlation degradation.

5.4.2 Correlation Tensor of the Composite Model

For the phase-evolved Bell state

$$|\psi(\theta)\rangle = \frac{1}{\sqrt{2}} \left(|01\rangle + e^{i\theta} |10\rangle \right), \tag{5.25}$$

the correlation tensor obtained previously is

$$T(\theta) = \begin{pmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & -1 \end{pmatrix}. \tag{5.26}$$

Using the tensor-contraction law derived in Section 5.3, the output tensor under local two-arm depolarization becomes

$$T_{\text{out}}(\theta, p_A, p_B) = \eta T(\theta), \qquad \eta = (1 - p_A)(1 - p_B). \tag{5.27}$$

Explicitly,

$$T_{\text{out}}(\theta, p_A, p_B) = \eta \begin{pmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & -1 \end{pmatrix}. \tag{5.28}$$

Therefore,

$$T_{xx}^{\text{out}} = \eta \cos\theta, \tag{5.29}$$
$$T_{xy}^{\text{out}} = -\eta \sin\theta, \tag{5.30}$$
$$T_{yx}^{\text{out}} = \eta \sin\theta, \tag{5.31}$$
$$T_{yy}^{\text{out}} = \eta \cos\theta, \tag{5.32}$$
$$T_{zz}^{\text{out}} = -\eta. \tag{5.33}$$

All remaining tensor components vanish within the reduced phase model considered here.

Equation (5.28) separates clearly the coherent and incoherent contributions of the effective fiber channel. The phase parameter θ rotates the transverse xy-correlation structure, while the depolarization factor η contracts the magnitude of the full bipartite tensor uniformly.

Thus, coherent phase evolution changes the orientation of the polarization correlations, whereas local depolarization suppresses their amplitude.

5.4.3 State Metrics Under the Composite Channel

The composite channel introduced above allows the simultaneous study of coherent phase mismatch and genuine non-unitary correlation degradation through quantitative state metrics.

Purity. For the composite phase–depolarization model, the global purity becomes

$$\gamma_{AB} = \mathrm{Tr}\left(\rho_{AB}^2\right) = \frac{1 + 3\eta^2}{4}, \tag{5.34}$$

where

$$\eta = (1 - p_A)(1 - p_B).$$

The purity depends only on the depolarization strength and is independent of the phase parameter θ. This reflects the fact that coherent unitary evolution preserves the eigenvalue spectrum of the density operator, while depolarization introduces genuine mixedness.

For maximally entangled Bell-state inputs, the reduced density operators remain maximally mixed:

$$\rho_A = \rho_B = \frac{I}{2}, \tag{5.35}$$

so that

$$\gamma_A = \gamma_B = \frac{1}{2}. \tag{5.36}$$

Bell-state fidelity. The fidelity with respect to the reference Bell state $|\Psi^+\rangle$ becomes

$$F_{\Psi^+} = \frac{1 + \eta \cos\theta}{2}. \tag{5.37}$$

This expression contains two distinct contributions.

The phase parameter θ modifies the relative orientation of the state with respect to the chosen Bell reference basis and may therefore reduce fidelity even in the absence of entanglement degradation. In contrast, the depolarization factor η reduces the coherence strength of the bipartite correlations themselves.

Consequently, reduced Bell-state fidelity does not necessarily imply reduced entanglement.

Concurrence. For the locally depolarized Bell-state family considered here, the concurrence is

$$C = \max\left[0, \frac{3\eta - 1}{2}\right]. \tag{5.38}$$

The concurrence is independent of the phase parameter θ, since local unitary transformations do not modify entanglement.

Therefore, coherent phase evolution changes the observed correlation structure while preserving bipartite entanglement, whereas local depolarization produces genuine entanglement degradation.

Entanglement disappears when

$$\eta = \frac{1}{3}. \tag{5.39}$$

For symmetric local depolarization,

$$p_A = p_B = p,$$

this condition becomes

$$(1 - p)^2 = \frac{1}{3}, \tag{5.40}$$

which gives

$$p = 1 - \frac{1}{\sqrt{3}} \approx 0.423. \tag{5.41}$$

CHSH nonlocality. Since arbitrary-axis correlation functions scale according to

$$E_{\text{out}} = \eta E_{\text{in}},$$

the maximal CHSH parameter also contracts linearly:

$$S_{\max}^{\text{out}} = \eta S_{\max}^{\text{in}}. \tag{5.42}$$

For a maximally entangled Bell state,

$$S_{\max}^{\text{in}} = 2\sqrt{2},$$

so that

$$S_{\max}^{\text{out}} = 2\sqrt{2}\,\eta. \tag{5.43}$$

Bell-inequality violation requires

$$S_{\max}^{\text{out}} > 2, \tag{5.44}$$

which implies

$$\eta > \frac{1}{\sqrt{2}}. \tag{5.45}$$

For symmetric local depolarization,

$$p < 1 - 2^{-1/4} \approx 0.159. \tag{5.46}$$

Therefore, CHSH nonlocality disappears before entanglement vanishes completely, showing that Bell-inequality violation is more fragile than concurrence under local depolarization.

5.4.4 Physical Interpretation

The composite channel developed in this section provides the first unified effective fiber model introduced in this work.

Within this framework, the phase parameter θ represents residual coherent birefringent evolution, while the depolarization parameters p_A and p_B describe effective incoherent polarization degradation occurring independently in the two propagation arms.

The correlation tensor acts as the bridge between the abstract density-operator description and experimentally accessible polarization measurements. Coherent phase evolution rotates the tensor structure relative to a fixed laboratory basis, whereas local depolarization contracts the tensor magnitude and progressively suppresses bipartite correlations.

The resulting model combines coherent polarization rotation and genuine non-unitary correlation degradation within a single analytically tractable framework.

This composite description constitutes the theoretical foundation for the numerical investigations developed in the following chapter, where the analytical predictions derived above are validated through simulation.

5.5 Statistical Composite Fiber Channel

The composite fiber model introduced in the previous section describes the simultaneous action of coherent phase evolution and local isotropic depolarization for a fixed phase parameter θ.

In realistic propagation scenarios, however, the relative phase accumulated during transmission may fluctuate statistically between different realizations. Consequently, the effective propagated state must combine both statistical phase averaging and local depolarizing degradation within a unified reduced description.

The purpose of this section is to extend the deterministic composite channel toward a statistical effective model capable of describing simultaneous coherent uncertainty and incoherent polarization degradation.

5.5.1 Ensemble-Averaged Composite Channel

Consider a statistical ensemble of phase realizations θ_k with associated probabilities p_k. The corresponding ensemble-averaged propagated state is

$$\rho_{AB}^{(\mathrm{ens})} = \sum_k p_k \left(\mathcal{D}_{p_A} \otimes \mathcal{D}_{p_B}\right) \left[U_{\theta_k} \rho_{AB}^{(\mathrm{in})} U_{\theta_k}^{\dagger}\right]. \tag{5.47}$$

Using the commutation property established previously, the local depolarizing channel may be moved outside the statistical average:

$$\rho_{AB}^{(\mathrm{ens})} = \left(\mathcal{D}_{p_A} \otimes \mathcal{D}_{p_B}\right) \left[\sum_k p_k U_{\theta_k} \rho_{AB}^{(\mathrm{in})} U_{\theta_k}^{\dagger}\right]. \tag{5.48}$$

Therefore, the statistical phase averaging and the isotropic local depolarization remain analytically separable within the effective model adopted in this work.

Following the notation introduced previously, define the phase-coherence parameter

$$\mu = \left\langle e^{i\theta} \right\rangle = \sum_k p_k e^{i\theta_k}. \tag{5.49}$$

Its real and imaginary parts satisfy

$$\mathrm{Re}(\mu) = \langle \cos\theta \rangle, \qquad \mathrm{Im}(\mu) = \langle \sin\theta \rangle. \tag{5.50}$$

The quantity $|\mu|$ therefore measures the residual phase coherence surviving the ensemble averaging process.

5.5.2 Correlation Tensor of the Statistical Composite Model

For the random-phase ensemble considered previously, the phase averaging modifies only the transverse correlation structure. Including local isotropic depolarization yields the effective tensor

$$T_{\mathrm{out}}^{(\mathrm{ens})} = \eta \begin{pmatrix} \mathrm{Re}(\mu) & -\mathrm{Im}(\mu) & 0 \\ \mathrm{Im}(\mu) & \mathrm{Re}(\mu) & 0 \\ 0 & 0 & -1 \end{pmatrix}. \tag{5.51}$$

Explicitly,

$$T_{xx}^{(\mathrm{ens})} = \eta\,\mathrm{Re}(\mu), \tag{5.52}$$
$$T_{xy}^{(\mathrm{ens})} = -\eta\,\mathrm{Im}(\mu), \tag{5.53}$$
$$T_{yx}^{(\mathrm{ens})} = \eta\,\mathrm{Im}(\mu), \tag{5.54}$$
$$T_{yy}^{(\mathrm{ens})} = \eta\,\mathrm{Re}(\mu), \tag{5.55}$$
$$T_{zz}^{(\mathrm{ens})} = -\eta. \tag{5.56}$$

All remaining tensor components vanish.

Equation (5.51) separates clearly the effects of the two physical mechanisms entering the model.

The statistical phase distribution modifies the transverse coherence structure through the parameter μ, while the local depolarizing channel contracts all bipartite correlations uniformly through the factor η.

An important consequence is that the longitudinal anticorrelation term T_{zz} is unaffected by phase averaging itself and is reduced only through local depolarization.

Therefore, phase fluctuations and isotropic depolarization leave distinct signatures on the correlation tensor structure.

5.5.3 State Metrics of the Statistical Composite Model

The statistical composite channel allows coherent uncertainty and incoherent polarization degradation to contribute simultaneously to the observable state metrics.

Purity. For the random-phase ensemble without local depolarization, the purity satisfies

$$\gamma\left(\rho_{AB}^{(\mathrm{ens})}\right) = \frac{1+|\mu|^2}{2}.$$

Including local isotropic depolarization yields

$$\gamma\left(\rho_{AB}^{(\mathrm{ens})}\right) = \frac{1+\eta^2\left(1+2|\mu|^2\right)}{4}. \tag{5.57}$$

The purity therefore decreases both when the phase coherence parameter $|\mu|$ is reduced and when the depolarization factor η decreases.

Bell-state fidelity. The Bell-state fidelity with respect to $|\Psi^+\rangle$ becomes

$$F_{\Psi^+} = \frac{1 + \eta \, \mathrm{Re}(\mu)}{2}. \tag{5.58}$$

This expression shows explicitly that the fidelity depends simultaneously on residual phase coherence and local depolarization strength.

Concurrence. For the statistical composite model, the concurrence becomes

$$C = \max \left[0, \eta|\mu| - \frac{1-\eta}{2} \right]. \tag{5.59}$$

Therefore, both statistical phase averaging and local depolarization contribute to entanglement degradation.

Phase averaging suppresses the coherence amplitude through $|\mu|$, while local depolarization contracts the bipartite correlations globally through the factor η.

CHSH nonlocality. Since the correlation tensor is globally contracted by η, the maximal CHSH parameter scales according to

$$S_{\max}^{(\mathrm{ens})} = 2\eta\sqrt{1 + |\mu|^2}. \tag{5.60}$$

Bell-inequality violation therefore requires

$$\eta\sqrt{1 + |\mu|^2} > 1. \tag{5.61}$$

Consequently, both reduced phase coherence and local depolarization suppress nonlocal quantum correlations.

5.5.4 Physical Interpretation

The statistical composite channel developed in this section provides the most general effective fiber model considered in the present work.

Within this framework, two physically distinct mechanisms contribute simultaneously to the degradation of bipartite polarization correlations.

The statistical phase distribution suppresses transverse coherence through ensemble averaging, reducing the effective phase-coherence parameter $|\mu|$. In contrast, isotropic local depolarization contracts the full correlation tensor uniformly through the factor η.

These mechanisms therefore affect the correlation structure differently. Phase averaging primarily modifies the coherent transverse interference structure, whereas local depolarization reduces the global magnitude of bipartite polarization correlations.

The resulting tensor structure provides a direct connection between microscopic propagation fluctuations and experimentally observable polarization-correlation measurements.

Most importantly, the model shows that coherent uncertainty and incoherent polarization degradation leave distinguishable signatures on entanglement, Bell-state fidelity, correlation tensors, and CHSH nonlocality.

This statistical composite framework constitutes the theoretical basis for the final numerical investigations developed in the following chapter.

Chapter 6

Numerical Experiments and Results

This chapter presents the numerical validation of the theoretical models developed in the previous chapters. The experiments are organized according to a progressive hierarchy of physical complexity, starting from deterministic unitary phase evolution and then extending toward stochastic and non-unitary propagation effects.
The objective is twofold: first, to verify the consistency between analytical predictions and numerical implementation; second, to investigate how different propagation mechanisms affect observable correlations, state purity, fidelity, concurrence, and nonlocality.
The numerical study follows the modeling hierarchy:

$$\begin{array}{c}\text{Deterministic Phase Evolution}\\ \downarrow \\ \text{Statistical Phase Fluctuations}\\ \downarrow \\ \text{Effective Depolarization}\\ \downarrow \\ \text{CHSH Nonlocality}\end{array}$$

6.1 Deterministic Phase Sweep

Objective and Physical Motivation

This experiment establishes the baseline propagation regime in which the optical fiber acts as an effective deterministic phase shifter preserving both the purity and the entanglement of the bipartite state.
The objective is to characterize how observable correlations and state-characterization quantities evolve under the reduced unitary fiber model introduced in Section 3.4.
This experiment represents the minimal physically relevant propagation scenario. The fiber is modeled exclusively through a deterministic differential phase shift acting on the polarization components of the transmitted photons, while all non-unitary effects are neglected.
Consequently, the experiment isolates the role of coherent phase evolution and establishes the baseline regime against which later noisy and depolarizing models will be compared.

Model and Analytical Predictions

The initial bipartite state is chosen as the Bell state

$$|\Psi^+\rangle = \frac{1}{\sqrt{2}}\left(|01\rangle + |10\rangle\right).$$

Under the reduced phase model, the fiber induces an effective relative phase shift between the polarization components, leading to the evolved state

$$|\psi(\theta)\rangle = \frac{1}{\sqrt{2}}\left(|01\rangle + e^{i\theta}|10\rangle\right). \tag{6.1}$$

The diagonal correlation observables considered in the simulation are

$$c_{xx} = \langle \sigma_x \otimes \sigma_x \rangle, \qquad c_{yy} = \langle \sigma_y \otimes \sigma_y \rangle, \qquad c_{zz} = \langle \sigma_z \otimes \sigma_z \rangle.$$

Direct analytical evaluation gives

$$c_{xx} = \cos\theta, \qquad c_{yy} = \cos\theta, \qquad c_{zz} = -1.$$

Since the evolution is generated by local unitary transformations, the global state remains pure:

$$\gamma(\rho(\theta)) = \mathrm{Tr}\left(\rho(\theta)^2\right) = 1.$$

The reduced density operators remain maximally mixed:

$$\rho_A(\theta) = \rho_B(\theta) = \frac{I}{2},$$

which implies

$$\gamma(\rho_A) = \gamma(\rho_B) = \frac{1}{2}.$$

The fidelity with respect to the reference Bell state $|\Psi^+\rangle$ is

$$F_{\Psi^+}(\rho(\theta)) = \frac{1+\cos\theta}{2}.$$

The concurrence remains constant:

$$C(\rho(\theta)) = 1,$$

demonstrating the invariance of bipartite entanglement under local unitary evolution.

The experiment therefore provides a controlled setting in which observable correlations evolve continuously with the phase parameter θ, while the underlying entanglement structure remains unchanged.

Numerical Results

A numerical sweep over the interval

$$\theta \in [0, 2\pi]$$

is performed in order to evaluate both correlation observables and state-characterization quantities throughout the phase evolution.
Figure 6.1 compares the numerical results obtained from the simulator with the corresponding analytical predictions for the diagonal correlation observables.

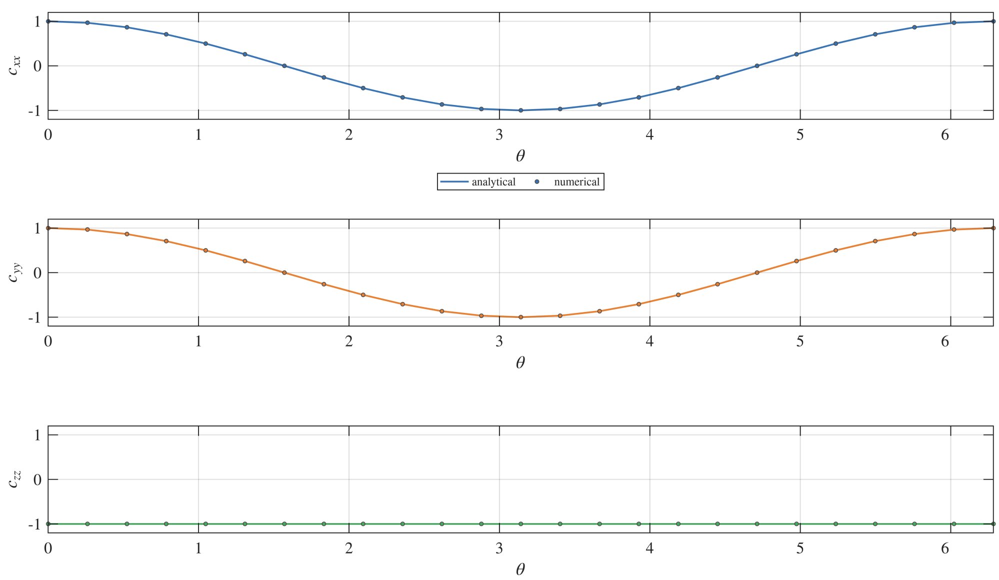


Figure 6.1: Numerical and analytical comparison for the correlation observables c_{xx}, c_{yy}, and c_{zz}. The agreement validates the implementation of the reduced phase model.

The numerical curves reproduce the expected cosine dependence for c_{xx} and c_{yy}, while c_{zz} remains constant at -1 over the entire propagation range.
The numerical and analytical curves are visually indistinguishable within numerical precision.

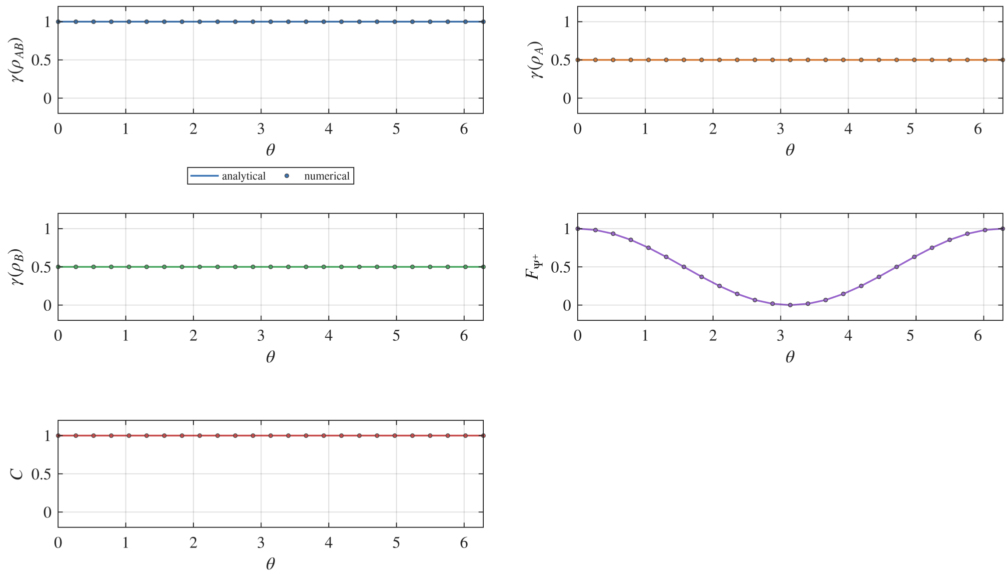


Figure 6.2: State-characterization quantities for the phase-evolved state. Global purity and concurrence remain constant, while fidelity varies with the relative phase θ.

Figure 6.2 shows the evolution of the state-characterization quantities.
The global purity remains equal to unity throughout the evolution, confirming that the propagated state remains pure. The reduced purities remain fixed at 1/2, while the concurrence remains equal to 1, indicating preservation of maximal entanglement.
In contrast, the fidelity with respect to the reference Bell state varies continuously with the phase parameter.

The full correlation tensor obtained numerically is shown in Figure 6.3.
The numerical results confirm the analytical tensor structure

$$T(\theta) = \begin{pmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & -1 \end{pmatrix}. \tag{6.2}$$

The non-zero off-diagonal components

$$T_{xy} = -\sin\theta, \qquad T_{yx} = \sin\theta$$

demonstrate that the phase evolution induces a rotation of the correlation structure within the transverse x–y plane.
This structure is not visible when considering only diagonal observables, but emerges naturally within the full tensor representation.

Discussion

The numerical results are in full agreement with the analytical predictions.

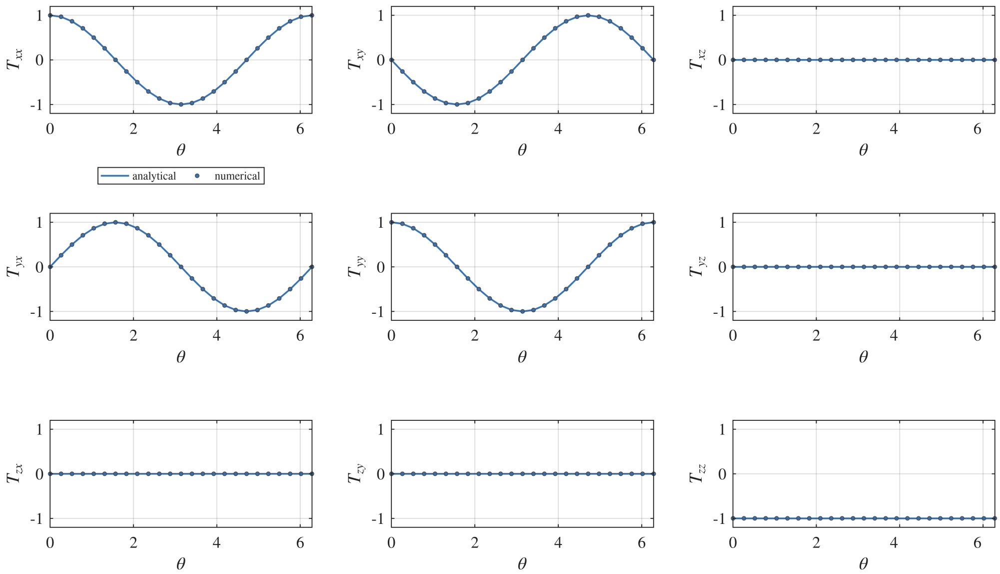


Figure 6.3: Full correlation tensor $T_{ij}(\theta)$ for the phase-evolved state. The off-diagonal tensor components reveal the rotation of the correlation structure in the transverse plane.

The experiment confirms that deterministic phase evolution modifies measurable two-photon correlations while preserving both the purity and the entanglement of the global state.
In particular, the transverse correlations c_{xx} and c_{yy} exhibit the expected cosine dependence on the phase parameter θ, whereas c_{zz} remains invariant.
At the same time, the concurrence remains constant despite the continuous variation of the fidelity with respect to the reference Bell state.
This distinction is physically significant.
The fidelity quantifies the overlap with a specific reference state and therefore depends explicitly on the relative phase. The concurrence, instead, measures the intrinsic entanglement content of the bipartite state and remains invariant under local unitary transformations.
Consequently, changes in fidelity do not necessarily imply degradation of entanglement.
The experiment therefore establishes a clear separation between:

- modifications of observable correlations induced by coherent phase evolution;
- genuine loss of entanglement caused by non-unitary processes.

This distinction will become essential in the interpretation of the later experiments involving statistical averaging and depolarization.

Physical Interpretation

The reduced phase model admits a direct physical interpretation in terms of fiber-induced birefringence.

In optical fibers, the horizontal and vertical polarization components generally propagate with different effective refractive indices. As a consequence, the two polarization modes accumulate different propagation phases.
After compensation of global polarization rotations, the dominant residual effect is a differential phase shift between the polarization components. Within the qubit representation, this residual evolution corresponds to a local unitary rotation around the Z-axis of the Bloch sphere.
The phase parameter θ therefore represents the effective relative phase accumulated during propagation.
The experiment shows that this mechanism modifies measurable correlations without degrading the entanglement of the transmitted state. The global quantum state remains pure, and the bipartite entanglement remains maximal throughout the evolution.
At the same time, the dependence of the transverse correlations on θ demonstrates that experimentally accessible observables remain highly sensitive to phase fluctuations.
This experiment therefore establishes the minimal effective model of coherent fiber propagation: a deterministic unitary transformation that continuously rotates the observable correlation structure while preserving the underlying entangled nature of the bipartite quantum state.

6.2 Random Phase Ensemble

Objective and Physical Motivation

The second experiment extends the deterministic phase model by introducing random phase realizations θ_k, modeling stochastic birefringence effects in realistic optical fibers.
Unlike the deterministic regime studied in Section 6.1, the system is no longer described by a single pure state, but by a statistical ensemble of phase-evolved realizations

$$\{|\psi(\theta_k)\rangle\}_{k=1}^{N},$$

together with the corresponding ensemble-averaged density operator.
The objective is to characterize how unresolved phase fluctuations modify observable correlations, global coherence, purity, fidelity, and entanglement.
This experiment therefore represents the transition from purely deterministic unitary evolution toward an effective mixed-state description arising from statistical averaging.

Model and Analytical Predictions

Each phase realization is described by the pure state

$$|\psi(\theta_k)\rangle = \frac{1}{\sqrt{2}}\left(|01\rangle + e^{i\theta_k}|10\rangle\right). \tag{6.3}$$

The ensemble-averaged state is then constructed as

$$\rho^{\text{ens}} = \frac{1}{N}\sum_{k=1}^{N} |\psi(\theta_k)\rangle\langle\psi(\theta_k)|. \tag{6.4}$$

Introducing the ensemble coherence parameter

$$\mu = \langle e^{i\theta}\rangle = \langle\cos\theta\rangle + i\langle\sin\theta\rangle, \tag{6.5}$$

the ensemble-averaged diagonal correlations become

$$c_{xx} = \text{Re}(\mu), \qquad c_{yy} = \text{Re}(\mu), \qquad c_{zz} = -1.$$

The global purity is

$$\gamma(\rho^{\text{ens}}) = \frac{1+|\mu|^2}{2}.$$

The reduced density operators remain maximally mixed:

$$\rho_A = \rho_B = \frac{I}{2},$$

which implies

$$\gamma(\rho_A) = \gamma(\rho_B) = \frac{1}{2}.$$

The fidelity with respect to the reference Bell state and the concurrence are given by

$$F_{\Psi^+} = \frac{1 + \mathrm{Re}(\mu)}{2}, \qquad C(\rho^{\mathrm{ens}}) = |\mu|.$$

The ensemble coherence parameter μ therefore provides a compact description of the entire propagation regime. Its real part determines observable transverse correlations and fidelity, while its magnitude controls the residual entanglement of the averaged state.

Numerical Results

Three representative phase distributions are considered:

- constant phase distribution;
- uniform phase distribution;
- Gaussian phase distribution.

For each case, the simulator evaluates:

- sampled phase statistics;
- pure-state correlation trends;
- ensemble-averaged correlations;
- purity, fidelity, and concurrence.

Figure 6.4 shows the constant-phase regime.

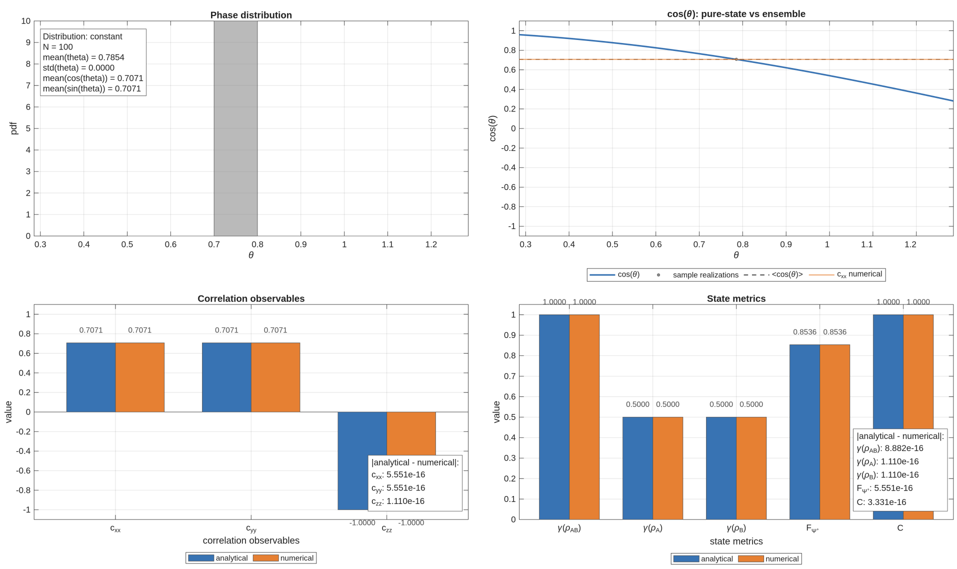


Figure 6.4: Constant phase distribution: the ensemble reduces to a deterministic realization and concurrence remains maximal.

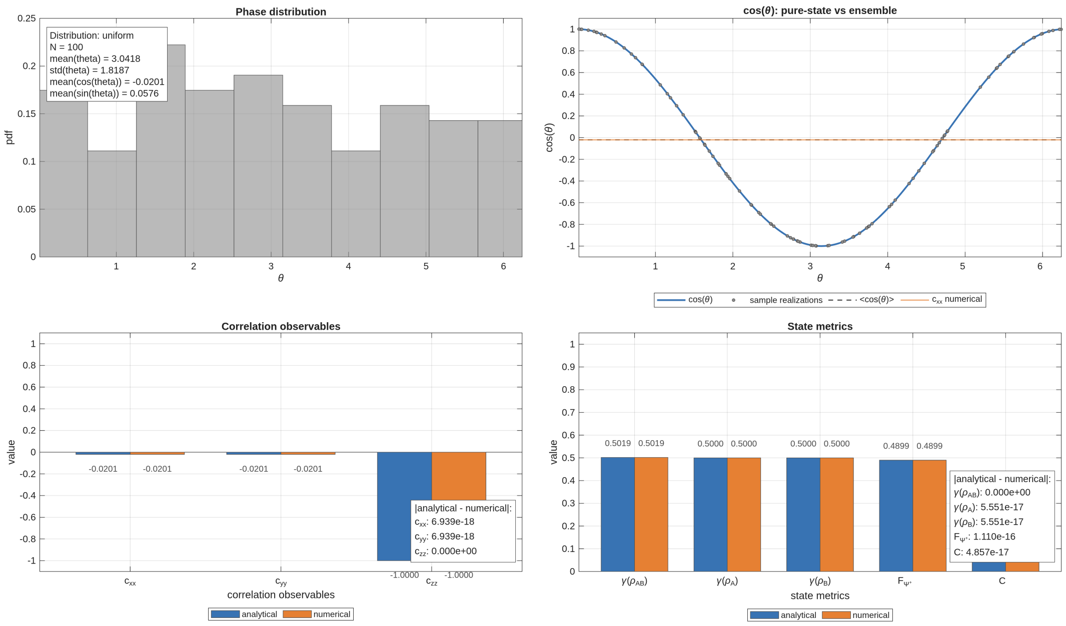


Figure 6.5: Uniform phase distribution: transverse correlations vanish, fidelity decreases, and concurrence tends to zero.

In this limit, all realizations coincide and the ensemble reproduces the deterministic behavior observed in Experiment 1.

Figure 6.5 shows the behavior corresponding to a uniform phase distribution.
The averaging process suppresses the transverse coherence terms, leading to vanishing ensemble correlations in the x–y plane and to a strong reduction of both fidelity and concurrence.

Figure 6.6 presents the intermediate regime generated by a Gaussian phase distribution.
In this case, residual coherence survives ensemble averaging, producing intermediate values of purity, fidelity, and concurrence.

The corresponding correlation tensor structures are shown in Figure 6.7.
As the number of realizations increases, the numerically reconstructed tensor components converge toward their analytical expectation values.
At the same time, tensor entries that are analytically zero remain numerically negligible throughout the simulation, confirming the internal consistency of the ensemble implementation.

Discussion

The numerical results reveal a progressive transition from deterministic coherent evolution to ensemble-induced decoherence.
The constant-phase regime reproduces the behavior observed in Experiment 1, where the ensemble effectively reduces to a single pure realization.
As the phase distribution broadens, the transverse coherence terms are progressively suppressed. Consequently, the ensemble coherence parameter satisfies

$$|\mu| \to 0,$$

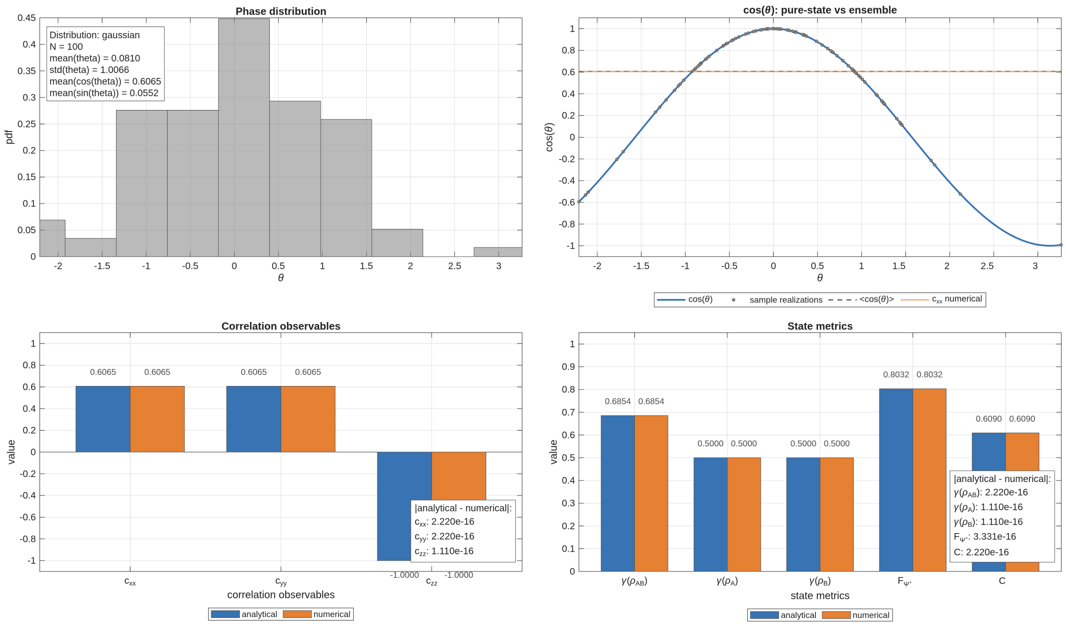
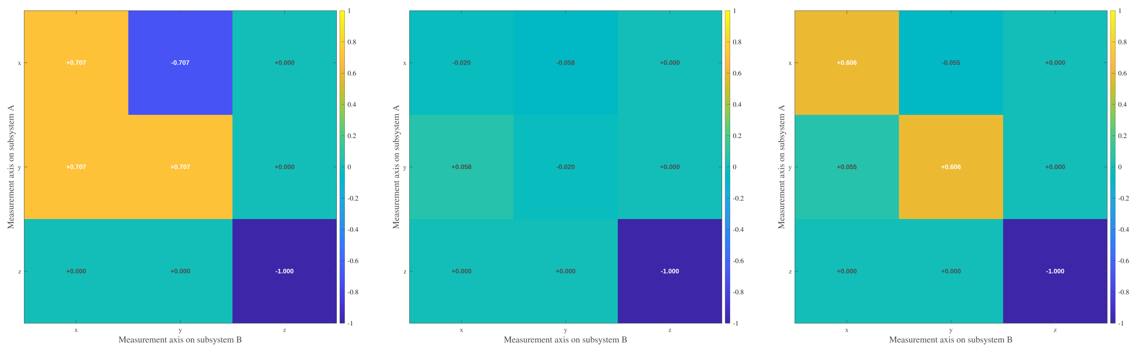


Figure 6.6: Gaussian phase distribution: partial coherence leads to intermediate values of purity and concurrence.



(a) Constant phase (b) Uniform distribution (c) Gaussian distribution

Figure 6.7: Correlation tensor snapshots for different phase distributions. Random phase averaging suppresses transverse coherence, progressively reducing the off-diagonal tensor components, while the longitudinal correlation $T_{zz} = -1$ remains invariant.

leading to the disappearance of coherent transverse correlations.
This transition is reflected directly in the state-characterization quantities:

- the global purity decreases from unity;
- the reduced purities remain fixed at 1/2;
- the fidelity decreases;
- the concurrence decreases continuously from 1 toward 0.

Unlike the deterministic phase evolution studied previously, the reduction of fidelity is now associated with a genuine degradation of entanglement.
The experiment therefore establishes an important conceptual distinction:

- deterministic phase evolution modifies observable correlations without destroying entanglement;
- unresolved statistical phase fluctuations generate effective mixedness and progressively suppress entanglement.

This mechanism constitutes the first non-unitary effect emerging naturally from the propagation model.

Physical Interpretation

Each individual realization of the ensemble remains a pure maximally entangled state. The loss of coherence arises only after averaging over unresolved phase fluctuations.
Physically, this situation models realistic fiber propagation conditions in which the accumulated birefringence-induced phase is not perfectly controlled and varies over time, frequency, or environmental conditions.
The ensemble coherence parameter

$$\mu = \langle e^{i\theta} \rangle$$

provides a unified physical interpretation of the propagation regime.
Its real part determines the observable transverse correlations and the fidelity with respect to the reference Bell state, while its magnitude quantifies the residual entanglement preserved after averaging.
As the phase fluctuations become broader, the ensemble loses its coherent phase structure. The corresponding density operator progressively approaches a mixed state, leading to a reduction of both purity and concurrence.
This experiment therefore captures the transition from ideal coherent propagation toward realistic noisy optical-fiber evolution, where stochastic birefringence effects induce effective decoherence and entanglement degradation.

6.3 Depolarizing Channel

Objective and Physical Motivation

This experiment investigates the degradation of a maximally entangled Bell state under an effective two-qubit depolarizing channel.
Unlike the previous experiments, which were based on deterministic or ensemble-averaged unitary evolution, the present model introduces a genuinely non-unitary transformation directly at the level of the density operator.
The objective is to characterize how depolarization affects:

- purity and mixedness;
- fidelity with respect to the Bell reference state;
- concurrence and entanglement degradation;
- observable correlations;
- the structure of the correlation tensor.

This experiment therefore establishes the first explicit decoherence model of the simulator and provides a baseline description of irreversible coherence loss in bipartite quantum propagation.

Model and Analytical Predictions

The initial state is the Bell state

$$|\Psi^+\rangle = \frac{|01\rangle + |10\rangle}{\sqrt{2}}, \qquad \rho_0 = |\Psi^+\rangle\langle\Psi^+|.$$

The depolarizing channel is defined as

$$\rho(p) = (1-p)\rho_0 + \frac{p}{4} I_4, \qquad p \in [0,1]. \tag{6.6}$$

The parameter p controls the strength of the depolarization process.
The limiting cases are

$$\rho(0) = \rho_0, \qquad \rho(1) = \frac{I_4}{4}.$$

Thus, the channel continuously interpolates between a pure maximally entangled Bell state and the maximally mixed two-qubit state.

For this family of states, the analytical predictions are

$$\gamma_{AB}(p) = 1 - \frac{3}{2}p + \frac{3}{4}p^2, \qquad \gamma_A = \gamma_B = \frac{1}{2},$$

$$F_{\Psi^+}(p) = 1 - \frac{3}{4}p, \qquad C(p) = \max\left(0, 1 - \frac{3}{2}p\right).$$

The concurrence vanishes at

$$p = \frac{2}{3},$$

marking the transition from entangled to separable states.

The action of the depolarizing channel on the correlation tensor corresponds to a uniform contraction of the Bell-state tensor:

$$T(p) = (1-p)T_{\Psi^+}.$$

Using the Bell-state tensor

$$T_{\Psi^+} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix},$$

one obtains

$$T(p) = \begin{pmatrix} 1-p & 0 & 0 \\ 0 & 1-p & 0 \\ 0 & 0 & -(1-p) \end{pmatrix}. \tag{6.7}$$

The corresponding diagonal correlations are therefore

$$c_{xx}(p) = 1-p, \qquad c_{yy}(p) = 1-p, \qquad c_{zz}(p) = -(1-p).$$

These expressions are consistent with the correlation-tensor formalism introduced in Section 3.5 and with the analytical derivations reported in Appendix **??**.

Numerical Results

The depolarization parameter p is sampled over the interval

$$p \in [0, 1],$$

and all state-characterization quantities and correlation observables are evaluated numerically and compared with the corresponding analytical predictions.
The numerical results show agreement at machine precision for all quantities considered in the simulation.
In particular:

- the Frobenius-norm error of the correlation tensor remains of order 10^{-16};
- the errors on purity, fidelity, concurrence, and diagonal correlations remain at numerical roundoff level.

Figure 6.8 summarizes the numerical evolution of the principal observables under the depolarizing channel.
The numerical and analytical curves are visually indistinguishable within numerical precision.
The global purity decreases monotonically from unity toward the maximally mixed limit, while the concurrence exhibits threshold behavior and vanishes at

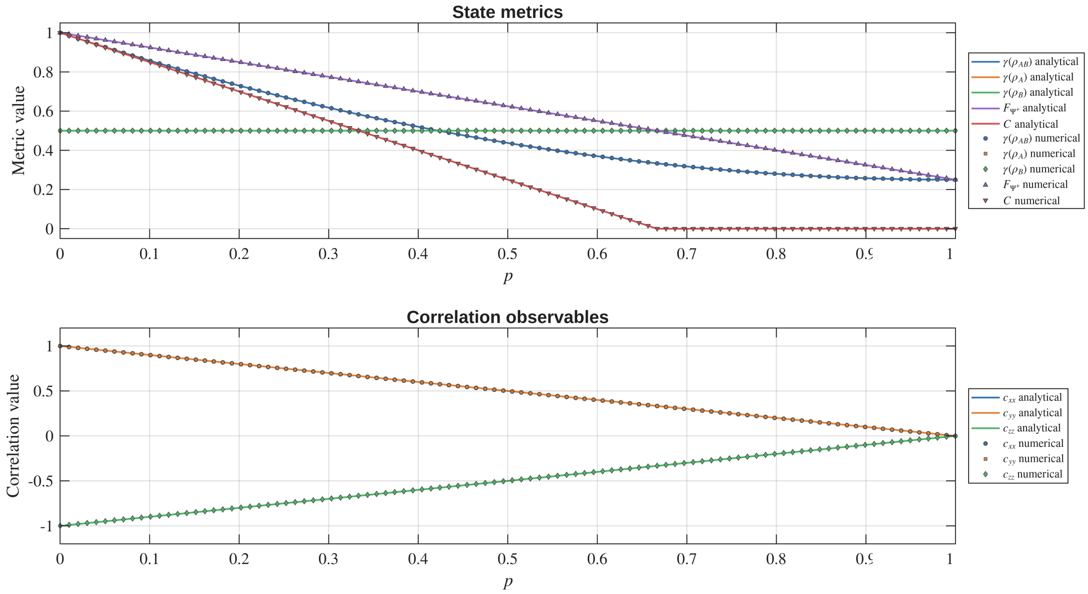


Figure 6.8: Numerical and analytical behavior of purity, fidelity, concurrence, and correlations under the depolarizing channel. Agreement is exact up to numerical precision.

$$p = \frac{2}{3}.$$

At the same time, the diagonal correlations decay linearly in magnitude, consistently with the tensor contraction law given in Eq. (6.7).

Discussion

This experiment validates the first explicit non-unitary extension of the simulator. Unlike the deterministic and ensemble-based phase models studied previously, the depolarizing channel introduces an isotropic loss of coherence directly at the level of the two-qubit density operator.
The numerical results show that the different observables exhibit distinct degradation behaviors:

- the global purity decreases smoothly over the full interval;
- the concurrence exhibits a sharp entanglement threshold;
- the observable correlations decay linearly;
- the full correlation tensor undergoes a uniform isotropic contraction.

The reduced density operators remain maximally mixed throughout the evolution:

$$\rho_A = \rho_B = \frac{I}{2}.$$

This behavior is consistent with the results of Experiment 6.2, where local mixedness also remained constant despite degradation of global coherence.

However, the physical origin of mixedness is fundamentally different in the two cases. In the random-phase ensemble model, mixedness arises from averaging over unresolved coherent realizations while preserving an underlying phase structure.
In contrast, the depolarizing channel suppresses the nontrivial two-qubit correlations isotropically by mixing the state directly with the maximally mixed operator $I_4/4$. The experiment therefore establishes a clear distinction between:

- decoherence emerging from statistical phase uncertainty;
- irreversible isotropic depolarization modeled through a CPTP channel.

Physical Interpretation

The depolarizing channel provides an effective phenomenological description of coherence loss when polarization information becomes partially inaccessible due to unresolved physical mechanisms.
Examples include:

- frequency-dependent birefringence;
- temporal walk-off;
- polarization-mode dispersion;
- unresolved environmental fluctuations.

The action of the channel progressively drives the Bell state toward the maximally mixed state, suppressing both observable correlations and entanglement.
Unlike the random-phase ensemble studied previously, the depolarizing model does not preserve any preferred phase structure. All nontrivial two-qubit correlations are uniformly reduced by the same contraction factor.
Although the depolarizing channel is not intended as a complete microscopic model of optical-fiber propagation, it provides a useful baseline description of decoherence and establishes a reference framework against which more realistic propagation models can later be compared.

6.4 CHSH Nonlocality

Objective and Physical Motivation

This experiment investigates the nonlocal correlations of the bipartite quantum state through the Clauser–Horne–Shimony–Holt (CHSH) parameter.
The analysis compares two physically distinct propagation mechanisms:

- deterministic phase evolution generated by local unitary transformations;
- depolarizing noise described by an effective two-qubit quantum channel.

The objective is to determine how these mechanisms affect the observable violation of the CHSH inequality and to distinguish between:

- apparent reductions of CHSH violation caused by measurement-axis misalignment;
- genuine loss of nonlocality caused by decoherence.

To this end, two different CHSH quantities are evaluated:

- S_{fixed}, computed using fixed laboratory measurement axes;
- S_{max}, computed from the correlation tensor and corresponding to the maximal CHSH value achievable after optimization over measurement settings.

This distinction is physically important because a decrease of S_{fixed} does not necessarily imply degradation of the intrinsic nonlocal properties of the quantum state.

Model and Analytical Predictions

The reference state is the Bell state

$$|\Psi^+\rangle = \frac{1}{\sqrt{2}} \left(|01\rangle + |10\rangle\right).$$

Two different state families are analyzed.

Phase-evolved states.
The first family is generated through deterministic local phase evolution:

$$|\psi(\theta)\rangle = \frac{1}{\sqrt{2}} \left(|01\rangle + e^{i\theta}|10\rangle\right), \tag{6.8}$$

obtained by applying the local unitary transformation

$$U_A(\theta) \otimes I_B.$$

Depolarized states.
The second family corresponds to the depolarizing channel:

$$\rho(p) = (1-p)\rho_0 + \frac{p}{4} I_4, \tag{6.9}$$

where

$$\rho_0 = |\Psi^+\rangle\langle\Psi^+|, \qquad p \in [0, 1].$$

The fixed CHSH measurement settings are chosen to be optimal for the Bell state at $\theta = 0$, whose correlation tensor is

$$T(0) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix}.$$

The local analyzer directions for subsystem A are

$$\mathbf{a}_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \qquad \mathbf{a}_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix},$$

while the analyzer directions for subsystem B are

$$\mathbf{b}_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \qquad \mathbf{b}_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}.$$

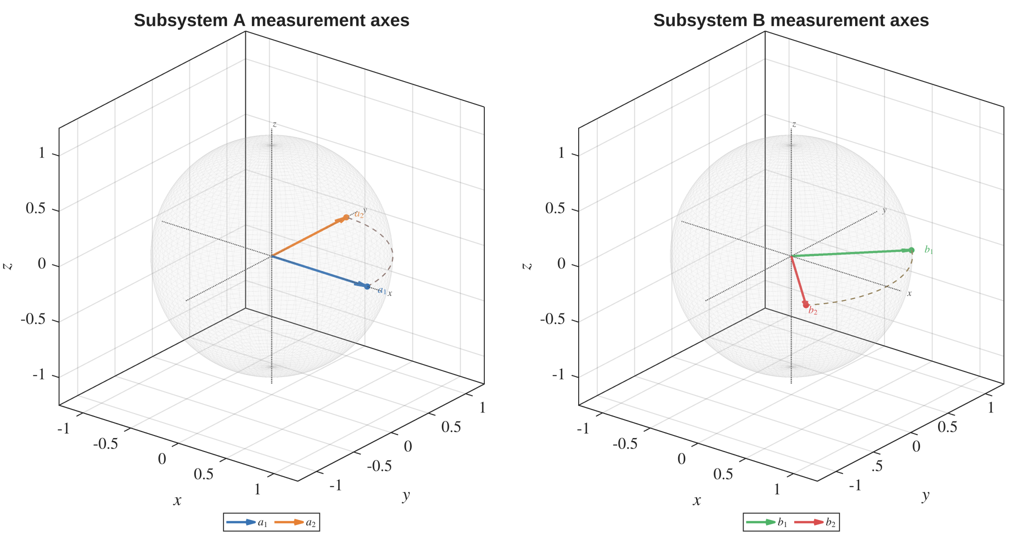


Figure 6.9: Fixed CHSH measurement axes represented on the Bloch spheres of subsystems A and B. The analyzer directions define the fixed laboratory configuration used to evaluate S_{fixed}.

The corresponding Bloch-sphere geometry is shown in Figure 6.9.

With these settings, S_{fixed} represents the CHSH value measured using a fixed analyzer configuration, whereas $S_{\max}$ represents the maximal violation achievable after optimization over local measurement axes.

For the phase-evolved Bell state, the correlation tensor becomes

$$T(\theta) = \begin{pmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & -1 \end{pmatrix}. \tag{6.10}$$

Using the fixed analyzer directions introduced above, the corresponding CHSH value is

$$S_{\text{fixed}}(\theta) = 2\sqrt{2}\cos\theta.$$

In the numerical plots, the quantity

$$|S_{\text{fixed}}|$$

is represented, since CHSH violation depends on the condition

$$|S| > 2.$$

The maximal CHSH value is instead independent of the phase parameter:

$$S_{\max} = 2\sqrt{2}.$$

This result reflects the invariance of nonlocality under local unitary transformations.

For the depolarizing channel, the correlation tensor is uniformly contracted:

$$T(p) = (1-p)T(0).$$

Consequently,

$$S_{\text{fixed}}(p) = 2\sqrt{2}(1-p), \qquad S_{\max}(p) = 2\sqrt{2}(1-p).$$

The CHSH violation threshold is determined from

$$S_{\max}(p) > 2,$$

which gives

$$p < p_{\text{CHSH}} = 1 - \frac{1}{\sqrt{2}} \simeq 0.2929.$$

Numerical Results

The numerical results are summarized in Figure 6.10.
In the deterministic phase-evolution regime, $|S_{\text{fixed}}|$ oscillates between zero and the Tsirelson bound.
This behavior originates from the relative rotation between the fixed laboratory analyzer axes and the phase-dependent correlation tensor.
At the same time, the maximal CHSH value remains constant:

$$S_{\max} = 2\sqrt{2},$$

demonstrating that the intrinsic nonlocality of the state is fully preserved.

In the depolarizing regime, the numerical and analytical curves overlap within numerical precision.

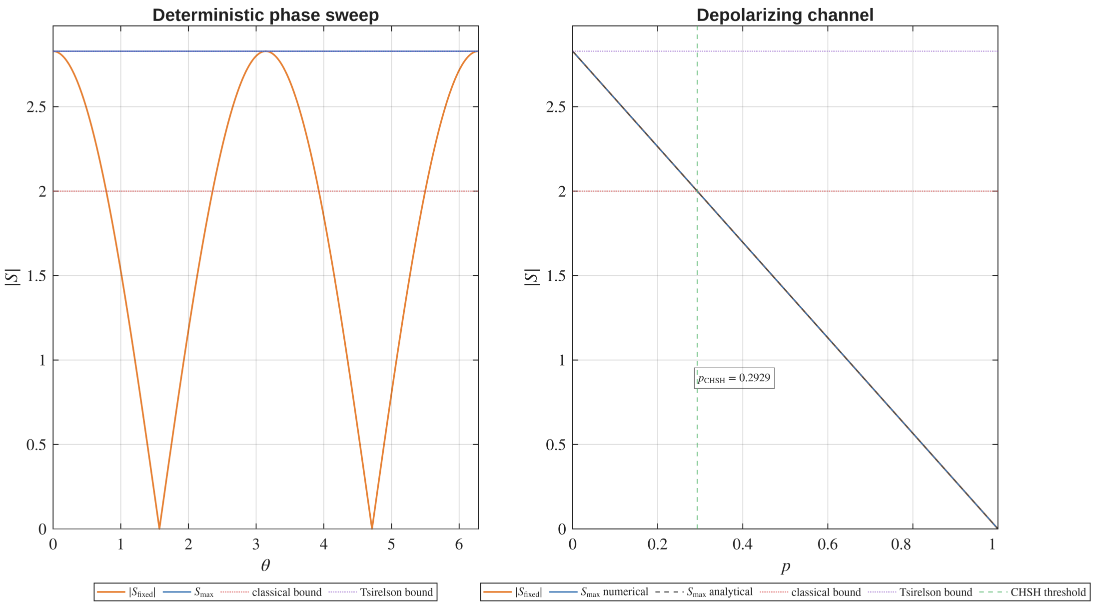


Figure 6.10: CHSH behavior under deterministic phase evolution and depolarizing noise. The left panel shows $|S_{\mathrm{fixed}}|$ and $S_{\max}$ as functions of the phase parameter θ. The right panel shows the corresponding quantities as functions of the depolarizing parameter p. The red dotted line indicates the classical CHSH bound $|S| = 2$, while the purple dotted line corresponds to the Tsirelson bound $2\sqrt{2}$.

Both $|S_{\mathrm{fixed}}|$ and $S_{\max}$ decrease linearly with the depolarization parameter p, crossing the classical CHSH bound at

$$p_{\mathrm{CHSH}} \simeq 0.2929.$$

Unlike the deterministic phase case, the reduction of CHSH violation now reflects a genuine loss of nonlocal correlation strength.

Discussion

The two propagation mechanisms considered in this experiment produce qualitatively different effects on CHSH nonlocality.
In the deterministic phase-evolution regime, the quantum state remains pure and maximally entangled for all values of the phase parameter θ.
Consequently, the oscillatory behavior of $|S_{\mathrm{fixed}}|$ does not indicate degradation of nonlocality. Instead, it reflects the fact that the fixed laboratory analyzer directions become progressively misaligned with the rotated correlation tensor.
This explains why $|S_{\mathrm{fixed}}|$ may fall below the classical CHSH threshold, or even vanish completely, while the optimized quantity $S_{\max}$ remains fixed at the Tsirelson bound.
The apparent loss of CHSH violation is therefore caused entirely by measurement-axis mismatch.

The depolarizing regime exhibits fundamentally different behavior.
The channel reduces the magnitude of the correlation tensor itself, producing a genuine contraction of the available nonlocal correlations.

As a consequence, no optimization of the analyzer directions can recover CHSH violation once the depolarization parameter exceeds the threshold

$$p_{\mathrm{CHSH}} = 1 - \frac{1}{\sqrt{2}}.$$

The experiment therefore establishes an important operational distinction:

- S_{fixed} is sensitive to both measurement misalignment and physical degradation;
- S_{max} isolates the maximal nonlocality intrinsically present in the quantum state.

Physical Interpretation

This experiment separates coherent fiber-induced transformations from incoherent decoherence mechanisms.
A deterministic relative phase acts as a local unitary transformation generated by birefringence-induced phase accumulation between polarization modes.
Such a mechanism rotates the correlation tensor but does not modify its singular values. Consequently, the intrinsic nonlocal resources of the Bell state remain fully preserved.
However, a fixed analyzer configuration may fail to detect the CHSH violation if the laboratory axes are no longer aligned with the rotated tensor.

Depolarization has a fundamentally different physical interpretation.
It models the effective loss of polarization coherence caused by unresolved degrees of freedom, stochastic fluctuations, frequency-dependent propagation effects, or polarization-mode dispersion.
In this case, the correlation tensor is not merely rotated but genuinely contracted, leading to irreversible degradation of the nonlocal correlations.
The comparison between S_{fixed} and S_{max} therefore provides an operational diagnostic tool:

- if S_{fixed} changes while S_{max} remains constant, the effect is compatible with coherent unitary evolution;
- if S_{max} decreases, the quantum state itself has lost nonlocal correlation strength.

For the quantum fiber simulator, this distinction is particularly important.
A fiber-induced phase shift may hide CHSH violation from fixed laboratory measurements without destroying the underlying entanglement resource, whereas depolarizing effects correspond to genuine physical degradation of the entangled photon pair.

Appendix A

Foundations of Bipartite Quantum States

A.1 Explicit Derivation of Reduced Density Operators and Comparison of Global States

This appendix provides an explicit derivation of the reduced density operators associated with the phase-evolved Bell state introduced in Chapter 3. It also illustrates an important conceptual point: identical local reduced states may correspond to physically different global states.

The discussion clarifies why local measurements alone are insufficient to characterize bipartite quantum systems and motivates the introduction of global quantities such as purity, correlation functions, and entanglement measures.

Phase-Evolved Bell State

We consider the two-qubit state

$$|\psi(\theta)\rangle = \frac{|01\rangle + e^{i\theta}|10\rangle}{\sqrt{2}}, \tag{A.1}$$

written in the computational basis

$$\{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}.$$

Its vector representation is

$$|\psi(\theta)\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ 1 \\ e^{i\theta} \\ 0 \end{pmatrix}. \tag{A.2}$$

The corresponding density operator is

$$\rho_{AB}(\theta) = |\psi(\theta)\rangle\langle\psi(\theta)|. \tag{A.3}$$

Expanding explicitly gives

$$\rho_{AB}(\theta) = \frac{1}{2}\begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & e^{-i\theta} & 0 \\ 0 & e^{i\theta} & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}. \tag{A.4}$$

Explicit Evaluation of the Partial Trace

The reduced density operator of subsystem A is obtained through the partial trace over subsystem B:

$$\rho_A(\theta) = \mathrm{Tr}_B\left(\rho_{AB}(\theta)\right). \tag{A.5}$$

Using the computational basis of subsystem B,

$$\{|0\rangle, |1\rangle\},$$

the partial trace can be written explicitly as

$$\rho_A(\theta) = \sum_{b=0}^{1} \left(I_A \otimes \langle b|\right) \rho_{AB}(\theta) \left(I_A \otimes |b\rangle\right). \tag{A.6}$$

We now evaluate the individual contributions.

The diagonal terms give

$$\begin{aligned} \mathrm{Tr}_B\left(|01\rangle\langle 01|\right) &= |0\rangle\langle 0|, \\ \mathrm{Tr}_B\left(|10\rangle\langle 10|\right) &= |1\rangle\langle 1|. \end{aligned} \tag{A.7}$$

The off-diagonal terms vanish:

$$\begin{aligned} \mathrm{Tr}_B\left(|01\rangle\langle 10|\right) &= 0, \\ \mathrm{Tr}_B\left(|10\rangle\langle 01|\right) &= 0, \end{aligned} \tag{A.8}$$

due to the orthogonality relation

$$\langle 0|1\rangle = 0.$$

Summing all contributions yields

$$\rho_A(\theta) = \frac{1}{2}|0\rangle\langle 0| + \frac{1}{2}|1\rangle\langle 1| = \frac{I}{2}. \tag{A.9}$$

By symmetry,

$$\rho_B(\theta) = \frac{I}{2}. \tag{A.10}$$

Thus, although the global state depends explicitly on the phase θ, the reduced states do not. The phase appears only in the off-diagonal coherence terms of the global density operator, which are eliminated by the partial trace.

In particular, for $\theta = 0$,

$$|\Psi^+\rangle = \frac{|01\rangle + |10\rangle}{\sqrt{2}}, \tag{A.11}$$

one obtains the same reduced states,

$$\rho_A = \rho_B = \frac{I}{2}. \tag{A.12}$$

Local Equivalence and Global Distinction

The previous result illustrates an important conceptual property of bipartite quantum systems.

Consider the classical mixture

$$\rho^{(\text{class})} = \frac{1}{2}|0\rangle\langle 0| + \frac{1}{2}|1\rangle\langle 1| = \frac{I}{2}. \tag{A.13}$$

At the local level,

$$\rho_A^{(\text{ent})} = \rho^{(\text{class})}, \tag{A.14}$$

and therefore, for any observable O_A,

$$\mathrm{Tr}\left(\rho^{(\text{class})} O_A\right) = \mathrm{Tr}\left(\rho_A^{(\text{ent})} O_A\right). \tag{A.15}$$

Hence, local measurements cannot distinguish classical mixtures from reductions of entangled states.

The distinction emerges only at the global level.
Consider the separable mixed state

$$\rho_{AB}^{(\text{sep})} = \frac{1}{2}|01\rangle\langle 01| + \frac{1}{2}|10\rangle\langle 10|. \tag{A.16}$$

Its reduced states are identical to those of the entangled Bell state:

$$\begin{aligned} \mathrm{Tr}_B\left(\rho_{AB}^{(\text{sep})}\right) &= \frac{I}{2}, \\ \mathrm{Tr}_A\left(\rho_{AB}^{(\text{sep})}\right) &= \frac{I}{2}. \end{aligned} \tag{A.17}$$

However, the separable state lacks the coherence terms present in the entangled density operator.

This difference becomes visible through global quantities.
For the entangled state,

$$\gamma\left(\rho_{AB}(\theta)\right) = 1, \tag{A.18}$$

whereas for the separable mixture,

$$\gamma\left(\rho_{AB}^{(\text{sep})}\right) = \frac{1}{2}. \tag{A.19}$$

A second distinction appears in bipartite correlations.

For the entangled phase-evolved state,

$$\langle \sigma_x \otimes \sigma_x \rangle = \cos\theta,$$
$$\langle \sigma_y \otimes \sigma_y \rangle = \cos\theta, \tag{A.20}$$

whereas for the separable mixture,

$$\langle \sigma_x \otimes \sigma_x \rangle = 0,$$
$$\langle \sigma_y \otimes \sigma_y \rangle = 0. \tag{A.21}$$

Thus, states with identical reduced density operators may differ fundamentally at the global level.
Reduced states encode all locally accessible information, but do not capture global coherence or bipartite entanglement.

A.2 Derivation of Fidelity for a Pure Reference State

This appendix derives the simplified expression for quantum fidelity when one of the two states is pure. The result is used throughout Chapter 4 in the evaluation of fidelity with respect to Bell states.

Beyond its algebraic simplification, this derivation also clarifies the operational interpretation of fidelity as a projection probability onto a target quantum state.

General Definition of Fidelity

Let ρ and σ be density operators acting on a finite-dimensional Hilbert space.
The general definition of quantum fidelity is

$$F(\rho,\sigma) = \left(\mathrm{Tr}\sqrt{\sqrt{\rho}\,\sigma\,\sqrt{\rho}}\right)^2. \tag{A.22}$$

In the present derivation, we consider the special case in which the reference state is pure:

$$\sigma = |\psi\rangle\langle\psi|, \tag{A.23}$$

with

$$\langle\psi|\psi\rangle = 1. \tag{A.24}$$

By construction, the operator σ is:

- Hermitian,
$$\sigma^\dagger = \sigma,$$
- positive semidefinite,
$$\langle\chi|\sigma|\chi\rangle \geq 0, \qquad \forall\,|\chi\rangle,$$
- normalized,
$$\mathrm{Tr}(\sigma) = 1.$$

Moreover, σ is a rank-one projector onto the one-dimensional subspace spanned by $|\psi\rangle$.

Reduction for a Pure Reference State

Substituting Eq. (A.23) into Eq. (A.22) gives

$$\sqrt{\rho}\,\sigma\,\sqrt{\rho} = \sqrt{\rho}\,|\psi\rangle\langle\psi|\,\sqrt{\rho}. \tag{A.25}$$

Define the vector

$$|\phi\rangle = \sqrt{\rho}\,|\psi\rangle. \tag{A.26}$$

Then Eq. (A.25) becomes

$$\sqrt{\rho}\,\sigma\,\sqrt{\rho} = |\phi\rangle\langle\phi|. \tag{A.27}$$

The operator $|\phi\rangle\langle\phi|$ is again rank one and positive semidefinite.

Spectral Structure of the Rank-One Operator

The only nonzero eigenvalue of the operator in Eq. (A.27) is

$$\lambda = \langle\phi|\phi\rangle. \tag{A.28}$$

Introducing the normalized vector

$$|\hat{\phi}\rangle = \frac{|\phi\rangle}{\|\phi\|}, \tag{A.29}$$

the spectral decomposition becomes

$$|\phi\rangle\langle\phi| = \lambda\,|\hat{\phi}\rangle\langle\hat{\phi}|. \tag{A.30}$$

The square root of the operator is therefore

$$\sqrt{|\phi\rangle\langle\phi|} = \sqrt{\lambda}\,|\hat{\phi}\rangle\langle\hat{\phi}|. \tag{A.31}$$

Taking the trace gives

$$\mathrm{Tr}\,\sqrt{|\phi\rangle\langle\phi|} = \sqrt{\lambda}. \tag{A.32}$$

Substituting into Eq. (A.22), one obtains

$$F(\rho,\sigma) = \lambda = \langle\phi|\phi\rangle. \tag{A.33}$$

Using Eq. (A.26),

$$\lambda = \langle\psi|\rho|\psi\rangle. \tag{A.34}$$

Final Expression and Interpretation

Combining the previous results, the fidelity simplifies to

$$F(\rho, |\psi\rangle\langle\psi|) = \langle\psi|\rho|\psi\rangle. \tag{A.35}$$

This is the expression used throughout Chapter 4 for fidelity with respect to Bell states.

The result admits a direct operational interpretation. Since

$$|\psi\rangle\langle\psi|$$

is the projector onto the state $|\psi\rangle$, the fidelity corresponds to the probability of obtaining the outcome $|\psi\rangle$ in a projective measurement performed on the state ρ.

Fidelity therefore quantifies the overlap between the physical state produced by the system and the chosen target reference state.

A.3 Derivations for Concurrence and Reduced Density Operators

This appendix derives the relations connecting concurrence, reduced density operators, and subsystem purity for pure bipartite two-qubit states.

The derivation clarifies how bipartite entanglement is encoded in the mixedness of the reduced subsystems, providing the theoretical foundation for the relations used throughout Chapter 4.

Coefficient-Matrix Representation

Consider a normalized two-qubit pure state written in the computational basis as

$$|\psi\rangle = a|00\rangle + b|01\rangle + c|10\rangle + d|11\rangle. \tag{A.36}$$

Introduce the coefficient matrix

$$A_\psi = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \tag{A.37}$$

so that the state can be expressed compactly as

$$|\psi\rangle = \sum_{i,j=0}^{1} (A_\psi)_{ij}\, |i\rangle \otimes |j\rangle. \tag{A.38}$$

This matrix representation provides a convenient bridge between the bipartite wavefunction and the reduced density operators.

Reduced Density Operators

The global density operator is

$$\rho_{AB} = |\psi\rangle\langle\psi|. \tag{A.39}$$

Expanding Eq. (A.38) yields

$$\rho_{AB} = \sum_{i,j,k,l} (A_\psi)_{ij}(A_\psi)^*_{kl} \, |i\rangle\langle k| \otimes |j\rangle\langle l|. \tag{A.40}$$

The reduced density operator on subsystem A is defined by the partial trace

$$\rho_A = \mathrm{Tr}_B(\rho_{AB}). \tag{A.41}$$

Applying the partial trace to Eq. (A.40) gives

$$\rho_A = \sum_{i,j,k,l} (A_\psi)_{ij}(A_\psi)^*_{kl} \left(\sum_v \langle v|j\rangle\langle l|v\rangle \right) |i\rangle\langle k|. \tag{A.42}$$

Using the orthogonality relation

$$\sum_v \langle v|j\rangle\langle l|v\rangle = \delta_{jl}, \tag{A.43}$$

one obtains

$$\rho_A = \sum_{i,k,j} (A_\psi)_{ij}(A_\psi)^*_{kj} \, |i\rangle\langle k|. \tag{A.44}$$

Therefore, in matrix form,

$$\rho_A = A_\psi A_\psi^\dagger. \tag{A.45}$$

Similarly,

$$\rho_B = A_\psi^\dagger A_\psi. \tag{A.46}$$

Pure-State Concurrence

For pure two-qubit states, concurrence is defined as

$$C(|\psi\rangle) = \left| \langle\psi|(\sigma_y \otimes \sigma_y)|\psi^*\rangle \right|. \tag{A.47}$$

Using the computational-basis representation,

$$|\psi\rangle = \begin{pmatrix} a \\ b \\ c \\ d \end{pmatrix}, \qquad |\psi^*\rangle = \begin{pmatrix} a^* \\ b^* \\ c^* \\ d^* \end{pmatrix}, \tag{A.48}$$

one finds

$$(\sigma_y \otimes \sigma_y)|\psi^*\rangle = \begin{pmatrix} -d^* \\ c^* \\ b^* \\ -a^* \end{pmatrix}. \tag{A.49}$$

Taking the inner product yields

$$\langle\psi|(\sigma_y \otimes \sigma_y)|\psi^*\rangle = 2(ad - bc)^*. \tag{A.50}$$

Therefore,

$$C(|\psi\rangle) = 2|ad - bc|. \tag{A.51}$$

Using

$$\det(A_\psi) = ad - bc, \tag{A.52}$$

the concurrence becomes

$$C(|\psi\rangle) = 2|\det(A_\psi)|. \tag{A.53}$$

Relation with Reduced Density Operators

Using Eq. (A.45),

$$\rho_A = A_\psi A_\psi^\dagger, \tag{A.54}$$

one obtains

$$\det(\rho_A) = \det(A_\psi)\det(A_\psi^\dagger) = |\det(A_\psi)|^2. \tag{A.55}$$

Substituting into Eq. (A.53) gives

$$C(|\psi\rangle) = 2\sqrt{\det(\rho_A)}. \tag{A.56}$$

This reproduces the relation introduced in Chapter 4 [see Eq. (4.27)].

An identical relation holds for subsystem B:

$$C(|\psi\rangle) = 2\sqrt{\det(\rho_B)}. \tag{A.57}$$

These expressions show that bipartite entanglement is directly encoded in the reduced density operators.

Relation with Purity

For a single-qubit density operator ρ, one has

$$\mathrm{Tr}(\rho) = 1, \tag{A.58}$$

and purity

$$\gamma(\rho) = \mathrm{Tr}(\rho^2). \tag{A.59}$$

For 2×2 density operators, the determinant satisfies

$$\det(\rho) = \frac{1 - \gamma(\rho)}{2}. \tag{A.60}$$

Substituting Eq. (A.60) into Eq. (4.27) yields

$$C(|\psi\rangle) = \sqrt{2\,(1 - \gamma(\rho_A))}. \tag{A.61}$$

Therefore, for pure bipartite states, the mixedness of the reduced subsystem provides a direct quantitative measure of entanglement.

Separable states produce pure reduced density operators and therefore zero concurrence, whereas maximally entangled states produce maximally mixed reduced states and maximal concurrence.

Appendix B

Basis Representation and Invariance Properties

B.1 Concurrence: Basis Dependence, Representation Dependence, and Invariance

This appendix clarifies the distinction between representation-dependent quantities and physically invariant quantities in quantum mechanics, with particular emphasis on the concurrence used throughout the present work.

A potential conceptual ambiguity arises from the standard definition of concurrence for mixed two-qubit states, since the construction explicitly involves the operator σ_y and complex conjugation in the computational basis. At first sight, this may suggest that concurrence depends on a preferred polarization axis or on a specific basis representation.
The purpose of this appendix is to show that this interpretation is incorrect, and that concurrence constitutes an intrinsic, basis-independent measure of bipartite entanglement.

Representation Dependence and Basis Transformations

In quantum mechanics, vectors and operators are represented through coordinates and matrices relative to a chosen basis.
For a single qubit, a generic pure state may be written as

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle.$$

The coefficients α and β depend on the chosen basis representation.
For example, switching from the computational basis

$$\{|0\rangle, |1\rangle\}$$

to the rotated basis

$$|+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}, \qquad |-\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}},$$

changes the numerical coordinates of the state vector, while leaving the physical quantum state unchanged.

Similarly, a density operator transforms under a basis change according to

$$\rho \to \rho' = U\rho U^\dagger, \tag{B.1}$$

where U is a unitary change-of-basis operator.
The individual matrix entries of ρ therefore depend on the chosen representation.
However, physically meaningful invariant quantities such as

$$\mathrm{Tr}(\rho), \qquad \mathrm{Tr}(\rho^2), \qquad \det(\rho), \qquad \mathrm{spec}(\rho)$$

remain unchanged under Eq. (B.1).

This distinction is fundamental:

- matrix elements and coordinate representations are generally basis dependent,
- intrinsic physical properties are basis independent.

Direction Dependence and Physical Observables

A separate concept is the dependence of observables on physical measurement directions.
Expectation values such as

$$\langle\sigma_x\rangle, \qquad \langle\sigma_y\rangle, \qquad \langle\sigma_z\rangle$$

correspond to measurements performed along different axes of the Bloch sphere.
These quantities are physically direction dependent, since changing the measurement axis changes the measured observable.

For instance,

$$\langle\sigma_z\rangle = \mathrm{Tr}(\rho\sigma_z)$$

measures polarization along the computational Z-axis, while

$$\langle\sigma_x\rangle = \mathrm{Tr}(\rho\sigma_x)$$

measures polarization along the transverse X-axis.

Direction dependence therefore refers to the physical choice of measurement axis, rather than to the mathematical choice of representation basis.
These two notions are related but conceptually distinct.

Representation Dependence of Pauli Matrices

The Pauli operators themselves admit different matrix representations depending on the chosen basis.
In the computational basis,

$$\sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}. \tag{B.2}$$

Under a basis transformation U, the matrix representation changes according to

$$\sigma_y \to \sigma_y' = U\sigma_y U^\dagger. \tag{B.3}$$

Consequently, the matrix entries appearing in Eq. (B.2) are representation dependent.

Importantly, this does not imply that the physical observable itself changes. Rather, only its coordinate representation changes.
This situation is directly analogous to ordinary vector coordinates in Euclidean geometry.

Concurrence and the Spin-Flip Construction

For mixed two-qubit states, concurrence is commonly expressed through the spin-flipped density operator

$$\tilde{\rho} = (\sigma_y \otimes \sigma_y)\rho^*(\sigma_y \otimes \sigma_y), \tag{B.4}$$

where ρ^* denotes complex conjugation in the computational basis.
The concurrence is then defined as

$$C(\rho) = \max\left(0, \lambda_1 - \lambda_2 - \lambda_3 - \lambda_4\right), \tag{B.5}$$

where λ_i are the ordered square roots of the eigenvalues of

$$\rho\tilde{\rho}.$$

At first sight, Eq. (B.4) may appear to privilege the Y-direction of the Bloch sphere because of the explicit presence of σ_y.
However, this interpretation is incorrect.
In this context, σ_y is not used as a measurement operator associated with the physical Y-axis. Instead, it appears because it realizes the invariant antisymmetric structure of two-dimensional spinor spaces.

Antisymmetric Structure of Two-Qubit States

Consider a generic pure two-qubit state

$$|\psi\rangle = a|00\rangle + b|01\rangle + c|10\rangle + d|11\rangle. \tag{B.6}$$

The coefficients may be organized into the matrix

$$\Psi = \begin{pmatrix} a & b \\ c & d \end{pmatrix}. \tag{B.7}$$

For pure states, concurrence reduces to

$$C(|\psi\rangle) = 2|ad - bc|. \tag{B.8}$$

The quantity

$$ad - bc$$

is precisely the determinant of the coefficient matrix Ψ:

$$\det(\Psi) = ad - bc. \tag{B.9}$$

This determinant originates from the antisymmetric bilinear form

$$\varepsilon = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \tag{B.10}$$

which satisfies

$$\sigma_y = -i\varepsilon. \tag{B.11}$$

Thus, the appearance of σ_y in the concurrence construction is not associated with a preferred physical polarization axis, but with the antisymmetric invariant structure of two-level quantum systems.

Basis Independence of Concurrence

An entanglement measure must remain invariant under local unitary transformations, since local polarization rotations cannot create or destroy entanglement.
For any local unitary evolution,

$$\rho \to (U_A \otimes U_B)\rho(U_A^\dagger \otimes U_B^\dagger), \tag{B.12}$$

the concurrence satisfies

$$C\left[(U_A \otimes U_B)\rho(U_A^\dagger \otimes U_B^\dagger)\right] = C(\rho). \tag{B.13}$$

This invariance is essential for the physical interpretation of concurrence as an intrinsic measure of bipartite entanglement.

Consequently:

- the matrix representation of the spin-flipped operator depends on the chosen basis,
- the intermediate construction in Eq. (B.4) is representation dependent,
- the final concurrence $C(\rho)$ is basis independent.

Manifestly Basis-Independent Formulations

For pure states, concurrence admits formulations that make the invariance more transparent.
Using the reduced density operator

$$\rho_A = \mathrm{Tr_B}\left(|\psi\rangle\langle\psi|\right),$$

one obtains

$$C(|\psi\rangle) = 2\sqrt{\det(\rho_A)}. \tag{B.14}$$

Equivalently,

$$C(|\psi\rangle) = \sqrt{2\,(1 - \mathrm{Tr}(\rho_A^2))}. \tag{B.15}$$

These expressions involve only invariant quantities such as determinants and traces, and therefore do not rely explicitly on any preferred basis representation.

More fundamentally, concurrence for mixed states is defined through the convex-roof construction

$$C(\rho) = \min_{\{p_k, |\psi_k\rangle\}} \sum_k p_k C(|\psi_k\rangle), \tag{B.16}$$

where

$$\rho = \sum_k p_k |\psi_k\rangle\langle\psi_k|. \tag{B.17}$$

This formulation is intrinsically basis independent.
The spin-flip formalism of Eq. (B.4) constitutes a special analytical construction that allows the explicit evaluation of the convex-roof optimization for two-qubit systems.

Physical Interpretation in Fiber-Based Quantum Channels

The distinction between representation-dependent quantities and invariant quantities plays a central role in the modeling strategy developed throughout the present work.

Local unitary transformations induced by fiber propagation,

$$U_A \otimes U_B,$$

may alter:

- local Bloch coordinates,
- Stokes parameters,
- tensor components,
- fixed-basis Bell-state fidelities,
- correlation functions evaluated along specific axes.

However, these transformations do not modify the intrinsic entanglement content of the bipartite state.
Consequently, the concurrence remains unchanged under coherent local polarization evolution.

By contrast, non-unitary effects such as ensemble averaging, decoherence, or polarization-mode dispersion generate effective mixed states and therefore reduce concurrence.
This distinction establishes the conceptual transition

$$\text{coherent local evolution} \rightarrow \text{effective decoherence} \rightarrow \text{entanglement degradation},$$

which constitutes one of the central physical mechanisms investigated in the simulator developed in this work. “‘

Appendix C

Quantum Channels and Effective Fiber Models

C.1 Global Depolarizing Channel Analytics

This appendix derives the analytical expressions used to validate the depolarizing-channel experiment in Section 6.3.
The reference state is the Bell state

$$|\Psi^+\rangle = \frac{1}{\sqrt{2}}\left(|01\rangle + |10\rangle\right),$$

with density operator

$$\rho_0 = |\Psi^+\rangle\langle\Psi^+|.$$

The depolarized family is defined as

$$\rho(p) = (1-p)\rho_0 + \frac{p}{4}I_4, \qquad p \in [0,1]. \tag{C.1}$$

Density matrix representation

In the computational basis

$$\{|00\rangle, |01\rangle, |10\rangle, |11\rangle\},$$

the Bell-state density operator is

$$\rho_0 = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

Substituting this expression into Eq. (C.1) gives

$$\rho(p) = \begin{pmatrix} \frac{p}{4} & 0 & 0 & 0 \\ 0 & \frac{1}{2} - \frac{p}{4} & \frac{1-p}{2} & 0 \\ 0 & \frac{1-p}{2} & \frac{1}{2} - \frac{p}{4} & 0 \\ 0 & 0 & 0 & \frac{p}{4} \end{pmatrix}.$$

Global purity

The global purity is defined as

$$\gamma_{AB}(p) = \mathrm{Tr}\left(\rho(p)^2\right).$$

Since ρ_0 is a rank-one projector and $I_4/4$ is proportional to the identity, the Bell-state direction remains an eigenvector of $\rho(p)$, with eigenvalue

$$\lambda_1 = 1 - \frac{3p}{4}.$$

The three orthogonal Bell-basis directions have eigenvalues

$$\lambda_2 = \lambda_3 = \lambda_4 = \frac{p}{4}.$$

Therefore,

$$\gamma_{AB}(p) = \left(1 - \frac{3p}{4}\right)^2 + 3\left(\frac{p}{4}\right)^2.$$

After simplification,

$$\gamma_{AB}(p) = 1 - \frac{3}{2}p + \frac{3}{4}p^2. \tag{C.2}$$

Reduced density operators

For the Bell state,

$$\mathrm{Tr}_B(\rho_0) = \mathrm{Tr}_A(\rho_0) = \frac{I_2}{2}.$$

Moreover,

$$\mathrm{Tr}_B\left(\frac{I_4}{4}\right) = \mathrm{Tr}_A\left(\frac{I_4}{4}\right) = \frac{I_2}{2}.$$

By linearity of the partial trace,

$$\rho_A(p) = \mathrm{Tr}_B(\rho(p)) = (1-p)\frac{I_2}{2} + p\frac{I_2}{2} = \frac{I_2}{2}.$$

Similarly,

$$\rho_B(p) = \frac{I_2}{2}.$$

The reduced purities are therefore

$$\gamma_A(p) = \gamma_B(p) = \mathrm{Tr}\left[\left(\frac{I_2}{2}\right)^2\right] = \frac{1}{2}. \tag{C.3}$$

Fidelity with respect to $|\Psi^+\rangle$

Since the reference state is pure, the fidelity is

$$F_{\Psi^+}(p) = \langle\Psi^+|\rho(p)|\Psi^+\rangle.$$

Substituting Eq. (C.1),

$$F_{\Psi^+}(p) = (1-p)\langle\Psi^+|\rho_0|\Psi^+\rangle + \frac{p}{4}\langle\Psi^+|I_4|\Psi^+\rangle.$$

Since

$$\langle\Psi^+|\rho_0|\Psi^+\rangle = 1, \qquad \langle\Psi^+|I_4|\Psi^+\rangle = 1,$$

one obtains

$$F_{\Psi^+}(p) = 1 - \frac{3}{4}p. \tag{C.4}$$

Concurrence

The density operator $\rho(p)$ is Bell-diagonal. For a Bell-diagonal two-qubit state, the concurrence is

$$C(p) = \max\left(0, 2\lambda_{\max} - 1\right),$$

where $\lambda_{\max}$ is the largest Bell-basis eigenvalue.
In the present case,

$$\lambda_{\max} = 1 - \frac{3p}{4}.$$

Therefore,

$$C(p) = \max\left(0, 1 - \frac{3p}{2}\right). \tag{C.5}$$

The concurrence vanishes when

$$1 - \frac{3p}{2} = 0,$$

which gives the separability threshold

$$p = \frac{2}{3}. \tag{C.6}$$

Correlation tensor

The Pauli correlations are defined as

$$c_{ij}(p) = \mathrm{Tr}\left[\rho(p)\left(\sigma_i \otimes \sigma_j\right)\right], \qquad i, j \in \{x, y, z\}.$$

Using linearity,

$$c_{ij}(p) = (1-p)\,\mathrm{Tr}\left[\rho_0\left(\sigma_i \otimes \sigma_j\right)\right] + \frac{p}{4}\,\mathrm{Tr}\left[I_4\left(\sigma_i \otimes \sigma_j\right)\right].$$

Since every Pauli product $\sigma_i \otimes \sigma_j$ with $i, j \in \{x, y, z\}$ is traceless,

$$\mathrm{Tr}\left[I_4 \left(\sigma_i \otimes \sigma_j\right)\right] = 0.$$

Thus,

$$c_{ij}(p) = (1-p)c_{ij}(0).$$

Equivalently, the full correlation tensor is uniformly contracted:

$$T(p) = (1-p)T_{\Psi^+}. \tag{C.7}$$

For the reference Bell state $|\Psi^+\rangle$,

$$T_{\Psi^+} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix}.$$

Therefore,

$$T(p) = \begin{pmatrix} 1-p & 0 & 0 \\ 0 & 1-p & 0 \\ 0 & 0 & -(1-p) \end{pmatrix}. \tag{C.8}$$

The diagonal correlations are

$$c_{xx}(p) = 1-p, \qquad c_{yy}(p) = 1-p, \qquad c_{zz}(p) = -(1-p). \tag{C.9}$$

Summary

Collecting the analytical expressions:

$$\boxed{\begin{aligned} \gamma_{AB}(p) &= 1 - \frac{3}{2}p + \frac{3}{4}p^2, \\ \gamma_A(p) = \gamma_B(p) &= \frac{1}{2}, \\ F_{\Psi^+}(p) &= 1 - \frac{3}{4}p, \\ C(p) &= \max\left(0, 1 - \frac{3}{2}p\right), \\ c_{xx}(p) &= 1-p, \qquad c_{yy}(p) = 1-p, \qquad c_{zz}(p) = -(1-p). \end{aligned}}$$

These expressions provide the analytical reference used for the numerical validation of the depolarizing-channel experiment.

C.2 Commutation of Local Phase Evolution and Isotropic Local Depolarization

This appendix establishes explicitly the commutation relation underlying the composite phase–depolarization fiber model introduced in Section 5.4.

In particular, it is shown that the reduced phase-evolution model considered in this work commutes with isotropic local depolarizing channels acting independently on the two subsystems. This property justifies the equivalence between applying local depolarization before or after coherent phase evolution within the isotropic approximation adopted in the present work.

Effective Channel Structure

The coherent phase evolution acting on subsystem A is represented through the unitary operator

$$U_\theta = E_\theta \otimes I, \tag{C.10}$$

where

$$E_\theta = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\theta} \end{pmatrix}. \tag{C.11}$$

The corresponding unitary channel is

$$\mathcal{U}_\theta(\rho) = U_\theta \rho U_\theta^\dagger. \tag{C.12}$$

Local isotropic depolarization acting independently on the two subsystems is described through

$$\mathcal{E}_{p_A,p_B} = \mathcal{D}_{p_A} \otimes \mathcal{D}_{p_B}, \tag{C.13}$$

where the single-qubit depolarizing channel satisfies

$$\mathcal{D}_p(I) = I, \tag{C.14}$$

and

$$\mathcal{D}_p(\sigma_i) = (1-p)\sigma_i, \qquad i \in \{x, y, z\}. \tag{C.15}$$

The objective is to prove that

$$\mathcal{E}_{p_A,p_B} \circ \mathcal{U}_\theta = \mathcal{U}_\theta \circ \mathcal{E}_{p_A,p_B}. \tag{C.16}$$

Pauli-Basis Expansion

Consider a generic two-qubit density operator expanded in the Pauli basis:

$$\rho = \frac{1}{4} \sum_{\mu,\nu=0}^{3} S_{\mu\nu}\, \sigma_\mu \otimes \sigma_\nu, \tag{C.17}$$

where

$$\sigma_0 = I,$$

and

$$\sigma_1 = \sigma_x, \qquad \sigma_2 = \sigma_y, \qquad \sigma_3 = \sigma_z.$$

Since both channels are linear, it is sufficient to study their action on generic Pauli-basis elements

$$\sigma_\mu \otimes \sigma_\nu.$$

Action of the Unitary Channel

The local phase unitary acts only on subsystem A. Therefore,

$$\mathcal{U}_\theta \left(\sigma_\mu \otimes \sigma_\nu \right) = \left(E_\theta \sigma_\mu E_\theta^\dagger \right) \otimes \sigma_\nu. \tag{C.18}$$

For $\mu = 0$, the identity operator is preserved trivially:

$$E_\theta I E_\theta^\dagger = I. \tag{C.19}$$

For $\mu \in \{1, 2, 3\}$, the transformed operator admits a Pauli-basis expansion of the form

$$E_\theta \sigma_\mu E_\theta^\dagger = a\sigma_0 + b\sigma_1 + c\sigma_2 + d\sigma_3. \tag{C.20}$$

However, unitary conjugation preserves tracelessness:

$$\mathrm{Tr} \left(E_\theta \sigma_\mu E_\theta^\dagger \right) = \mathrm{Tr}(\sigma_\mu) = 0, \qquad \mu \in \{1, 2, 3\}. \tag{C.21}$$

Since

$$\mathrm{Tr}(\sigma_0) = 2, \qquad \mathrm{Tr}(\sigma_i) = 0 \quad (i = 1, 2, 3),$$

Eq. (C.20) implies

$$a = 0.$$

Therefore,

$$E_\theta \sigma_\mu E_\theta^\dagger = b\sigma_1 + c\sigma_2 + d\sigma_3, \qquad \mu \in \{1, 2, 3\}. \tag{C.22}$$

The unitary evolution therefore preserves the traceless Pauli subspace.

Comparison of the Two Channel Orders

Consider first the ordering

$$\mathcal{E}_{p_A, p_B} \circ \mathcal{U}_\theta. \tag{C.23}$$

Acting on a generic basis element gives

$$\begin{aligned}&\mathcal{E}_{p_A,p_B}\left[\mathcal{U}_\theta\left(\sigma_\mu\otimes\sigma_\nu\right)\right]\\&=\mathcal{D}_{p_A}\left(E_\theta\sigma_\mu E_\theta^\dagger\right)\otimes\mathcal{D}_{p_B}\left(\sigma_\nu\right).\end{aligned}\tag{C.24}$$

Using Eq. (C.22), the first factor contains only traceless Pauli operators. Consequently,

$$\begin{aligned}&\mathcal{D}_{p_A}\left(E_\theta\sigma_\mu E_\theta^\dagger\right)\\&=(1-p_A)\left(E_\theta\sigma_\mu E_\theta^\dagger\right),\qquad \mu\in\{1,2,3\}.\end{aligned}\tag{C.25}$$

Now consider the opposite ordering:

$$\mathcal{U}_\theta\circ\mathcal{E}_{p_A,p_B}.\tag{C.26}$$

Its action gives

$$\begin{aligned}&\mathcal{U}_\theta\left[\mathcal{E}_{p_A,p_B}\left(\sigma_\mu\otimes\sigma_\nu\right)\right]\\&=E_\theta\mathcal{D}_{p_A}(\sigma_\mu)E_\theta^\dagger\otimes\mathcal{D}_{p_B}(\sigma_\nu).\end{aligned}\tag{C.27}$$

Using Eq. (C.15),

$$\begin{aligned}&E_\theta\mathcal{D}_{p_A}(\sigma_\mu)E_\theta^\dagger\\&=(1-p_A)E_\theta\sigma_\mu E_\theta^\dagger.\end{aligned}\tag{C.28}$$

Comparing Eqs. (C.25) and (C.28) shows that the two orderings coincide.

Therefore,

$$\mathcal{E}_{p_A,p_B}\circ\mathcal{U}_\theta=\mathcal{U}_\theta\circ\mathcal{E}_{p_A,p_B}.\tag{C.29}$$

Physical Interpretation

The commutation property derived above follows from two structural features of the effective model:

- the isotropic depolarizing channel acts uniformly on the traceless Pauli subspace;
- unitary conjugation preserves the traceless structure of Pauli operators.

Consequently, coherent phase evolution rotates the polarization-correlation structure without modifying its magnitude, while isotropic local depolarization contracts the correlation amplitudes uniformly without introducing preferred directions in polarization space.

Within the isotropic approximation adopted in the present work, the order of coherent phase evolution and local depolarization is therefore irrelevant.

This property considerably simplifies the analytical structure of the composite phase–depolarization fiber model introduced in Section 5.4, since coherent phase evolution and isotropic local depolarization may be treated independently while remaining mathematically equivalent under composition.

C.3 Statistical Composite Fiber-Channel Analytics

This appendix derives the analytical expressions associated with the statistical composite fiber model introduced in Section 5.5.

The model combines:

- statistical phase averaging,
- local isotropic depolarization acting independently on the two propagation arms.

The reference Bell state is

$$|\Psi^+\rangle = \frac{1}{\sqrt{2}} (|01\rangle + |10\rangle) ,$$

with density operator

$$\rho_0 = |\Psi^+\rangle\langle\Psi^+|.$$

The phase-evolved pure state is

$$|\psi(\theta)\rangle = \frac{1}{\sqrt{2}} \left(|01\rangle + e^{i\theta}|10\rangle\right) . \tag{C.30}$$

Define the phase-coherence parameter

$$\mu = \left\langle e^{i\theta} \right\rangle = \sum_k p_k e^{i\theta_k}, \tag{C.31}$$

and the depolarization contraction factor

$$\eta = (1 - p_A)(1 - p_B). \tag{C.32}$$

Ensemble density operator

The ensemble-averaged density operator is

$$\rho^{(\text{ens})} = \sum_k p_k |\psi(\theta_k)\rangle\langle\psi(\theta_k)|. \tag{C.33}$$

Using Eq. (C.30), one obtains

$$\rho^{(\mathrm{ens})} = \frac{1}{2}\begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & \mu^* & 0 \\ 0 & \mu & 1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}. \tag{C.34}$$

Including local isotropic depolarization contracts the bipartite correlations by the factor η. The resulting effective state is

$$\rho_\eta^{(\mathrm{ens})} = \frac{1}{4}\left[I \otimes I + \sum_{i,j} T_{ij}^{(\mathrm{ens})} \sigma_i \otimes \sigma_j \right], \tag{C.35}$$

where the tensor components are derived below.

Correlation tensor

For the ensemble state without local depolarization, the correlation tensor is

$$T^{(\mathrm{ens})} = \begin{pmatrix} \mathrm{Re}(\mu) & -\mathrm{Im}(\mu) & 0 \\ \mathrm{Im}(\mu) & \mathrm{Re}(\mu) & 0 \\ 0 & 0 & -1 \end{pmatrix}. \tag{C.36}$$

Under local isotropic depolarization,

$$T_\eta^{(\mathrm{ens})} = \eta T^{(\mathrm{ens})}, \tag{C.37}$$

thus

$$T_\eta^{(\mathrm{ens})} = \eta \begin{pmatrix} \mathrm{Re}(\mu) & -\mathrm{Im}(\mu) & 0 \\ \mathrm{Im}(\mu) & \mathrm{Re}(\mu) & 0 \\ 0 & 0 & -1 \end{pmatrix}. \tag{C.38}$$

The tensor components are therefore

$$T_{xx} = \eta\,\mathrm{Re}(\mu), \tag{C.39}$$
$$T_{xy} = -\eta\,\mathrm{Im}(\mu), \tag{C.40}$$
$$T_{yx} = \eta\,\mathrm{Im}(\mu), \tag{C.41}$$
$$T_{yy} = \eta\,\mathrm{Re}(\mu), \tag{C.42}$$
$$T_{zz} = -\eta. \tag{C.43}$$

All remaining tensor components vanish.

Global purity

The global purity is

$$\gamma\left(\rho_{AB}^{(\mathrm{ens})}\right) = \mathrm{Tr}\left[\left(\rho_{AB}^{(\mathrm{ens})}\right)^2\right]. \tag{C.44}$$

Using the Pauli-basis representation,

$$\rho = \frac{1}{4}\left[I \otimes I + \sum_{i,j} T_{ij}\sigma_i \otimes \sigma_j\right], \tag{C.45}$$

together with Pauli orthogonality,

$$\mathrm{Tr}\left[(\sigma_i \otimes \sigma_j)(\sigma_k \otimes \sigma_l)\right] = 4\delta_{ik}\delta_{jl}, \tag{C.46}$$

one obtains

$$\gamma = \frac{1}{4}\left[1 + \sum_{i,j} T_{ij}^2\right]. \tag{C.47}$$

Substituting Eq. (C.38),

$$\sum_{i,j} T_{ij}^2 = \eta^2\left[2|\mu|^2 + 1\right]. \tag{C.48}$$

Therefore,

$$\boxed{\gamma\left(\rho_{AB}^{(\mathrm{ens})}\right) = \frac{1 + \eta^2(1 + 2|\mu|^2)}{4}} \tag{C.49}$$

For $\eta = 1$, one recovers

$$\gamma\left(\rho_{AB}^{(\mathrm{ens})}\right) = \frac{1 + |\mu|^2}{2}. \tag{C.50}$$

Reduced density operators

The reduced density operators remain maximally mixed:

$$\rho_A = \rho_B = \frac{I_2}{2}. \tag{C.51}$$

Therefore,

$$\boxed{\gamma_A = \gamma_B = \frac{1}{2}}. \tag{C.52}$$

Fidelity with respect to $|\Psi^+\rangle$

The Bell-state fidelity is

$$F_{\Psi^+} = \langle\Psi^+|\rho_\eta^{(\mathrm{ens})}|\Psi^+\rangle. \tag{C.53}$$

Using the ensemble density operator,

$$F_{\Psi^+} = \frac{1}{2}\left[1 + \eta\,\mathrm{Re}(\mu)\right]. \tag{C.54}$$

Therefore,

$$\boxed{F_{\Psi^+} = \frac{1 + \eta\,\mathrm{Re}(\mu)}{2}} \tag{C.55}$$

For $\eta = 1$, this reduces to

$$F_{\Psi^+} = \frac{1 + \mathrm{Re}(\mu)}{2}. \tag{C.56}$$

Concurrence

The ensemble density operator belongs to the class of X-states. For a two-qubit X-state,

$$C = 2\max\left(0, |\rho_{23}| - \sqrt{\rho_{11}\rho_{44}}\right). \tag{C.57}$$

For the present model,

$$\rho_{23} = \frac{\eta\mu}{2}, \tag{C.58}$$

while the depolarizing contribution generates

$$\rho_{11} = \rho_{44} = \frac{1-\eta}{4}. \tag{C.59}$$

Therefore,

$$\begin{aligned} C &= 2\max\left[0, \frac{\eta|\mu|}{2} - \frac{1-\eta}{4}\right] \\ &= \max\left[0, \eta|\mu| - \frac{1-\eta}{2}\right]. \end{aligned} \tag{C.60}$$

Hence,

$$\boxed{C = \max\left[0, \eta|\mu| - \frac{1-\eta}{2}\right]} \tag{C.61}$$

CHSH nonlocality

The maximal CHSH parameter is determined by the Horodecki criterion:

$$S_{\max} = 2\sqrt{\lambda_1 + \lambda_2}, \tag{C.62}$$

where λ_1, λ_2 are the two largest eigenvalues of

$$T^T T.$$

Using Eq. (C.38),

$$T^T T = \eta^2 \begin{pmatrix} |\mu|^2 & 0 & 0 \\ 0 & |\mu|^2 & 0 \\ 0 & 0 & 1 \end{pmatrix}. \tag{C.63}$$

The two largest eigenvalues are therefore

$$\lambda_1 = \eta^2, \qquad \lambda_2 = \eta^2|\mu|^2.$$

Thus,

$$\boxed{S_{\max} = 2\eta\sqrt{1+|\mu|^2}} \tag{C.64}$$

Bell-inequality violation requires

$$S_{\max} > 2, \tag{C.65}$$

which gives

$$\boxed{\eta\sqrt{1+|\mu|^2} > 1} \tag{C.66}$$

Summary

Collecting the analytical expressions:

$$\boxed{\begin{aligned}
T_{xx} &= \eta\,\mathrm{Re}(\mu), \\
T_{xy} &= -\eta\,\mathrm{Im}(\mu), \\
T_{yx} &= \eta\,\mathrm{Im}(\mu), \\
T_{yy} &= \eta\,\mathrm{Re}(\mu), \\
T_{zz} &= -\eta, \\
\gamma\left(\rho_{AB}^{(\mathrm{ens})}\right) &= \frac{1+\eta^2(1+2|\mu|^2)}{4}, \\
\gamma_A = \gamma_B &= \frac{1}{2}, \\
F_{\Psi^+} &= \frac{1+\eta\,\mathrm{Re}(\mu)}{2}, \\
C &= \max\left[0, \eta|\mu| - \frac{1-\eta}{2}\right], \\
S_{\max} &= 2\eta\sqrt{1+|\mu|^2}.
\end{aligned}}$$

These expressions provide the analytical reference for the statistical composite fiber-channel model introduced in Section 5.5.

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